

Chapter 1: Foundations of Intelligence

Information, Analysis, and Decision Advantage

Companion Workbook to the Instructional Video

How to Use This Workbook

This workbook is the written companion to the instructional video on intelligence fundamentals. It is built to work three ways.

As a follow-along guide. Watch the video first, then read each section here. The video introduces the concepts. The workbook expands them with examples, tables, and diagrams that are hard to absorb at video speed.

As a standalone textbook chapter. If you never watch the video, you can still learn the material. Nothing in this chapter depends on having seen the recording.

As a practice manual. Roughly half of this workbook is exercises. Reading about intelligence does not make you an analyst. Producing intelligence does. Every lab in this book asks you to write something, judge something, or defend something.

A note on the audience. This chapter assumes you are comfortable thinking in systems and following a technical process, but it does not assume you have worked in intelligence, law enforcement, the military, or a security operations center. Where a term comes from a specialized field, it gets defined in plain language the first time it appears and again in the Key Terms section.

Time estimate for the full chapter:

Component	Estimated time
Video	45 to 60 minutes
Reading material	3 to 4 hours
Review questions	1 hour
Critical thinking exercises	1 to 2 hours
Labs 1 through 5	8 to 12 hours total
Capstone assignment	4 to 6 hours
Total	18 to 26 hours

Materials you will need: a notebook or word processor, internet access for open-source research in Labs 3 and 5, and a printer if you prefer to complete worksheets by hand. No specialized software is required. No paid tools are required.

Learning Objectives

By the end of this chapter, you should be able to do all of the following.

Knowledge objectives (you can explain it):

1. Define information and intelligence, and explain the specific differences that separate them.
2. Explain why raw data has no analytic value until it is placed in context.
3. Name and describe the phases of the intelligence lifecycle in the correct order.
4. Explain what happens when a phase of the lifecycle is skipped or performed poorly.
5. Identify the major intelligence collection disciplines and describe what each one collects.
6. Distinguish strategic, tactical, and operational intelligence by time horizon, audience, and purpose.
7. Describe how intelligence practice differs between government and private sector settings.
8. Explain the relationship between competitive intelligence, business intelligence, and criminal intelligence.
9. Describe the ethical and legal boundaries that separate lawful intelligence work from espionage, stalking, and unfair competition.

Application objectives (you can do it):

1. Classify a given item as data, information, or intelligence, and justify the classification.
2. Convert a set of raw reports into a written intelligence product with a judgment and a confidence level.
3. Map a real-world scenario onto the full intelligence lifecycle, phase by phase.
4. Select the appropriate intelligence discipline or combination of disciplines for a stated requirement.
5. Write a decision-ready intelligence summary that a busy executive can act on in under two minutes.
6. Assign and defend a confidence level for an analytic judgment.

Analysis objectives (you can judge it):

1. Evaluate the reliability of a source and the credibility of a report as separate questions.
2. Recognize common analytic failures, including confirmation bias, mirror imaging, and mistaking activity for intent.
3. Critique an intelligence product and explain what would make it more useful to the decision maker.

Self-check before you continue: if you cannot currently explain the difference between "the server logged 4,000 failed logins last night" and "an external actor is conducting a password spraying campaign against our VPN and will likely succeed within days unless we enable lockout," you are in the right place. That gap is the subject of this chapter.

Part 1: Reading Material

1.1 Information Versus Intelligence

1.1.1 The Core Distinction

Most people use the words *information* and *intelligence* as if they mean the same thing. They do not, and the difference is the single most important idea in this chapter.

Information is a collection of facts. It exists whether or not anyone needs it, uses it, or understands it. Information is neutral. It sits in a log file, a filing cabinet, a news broadcast, or a database until somebody picks it up.

Intelligence is information that has been collected on purpose, evaluated for reliability, analyzed in context, and delivered to a specific decision maker in time to affect a decision.

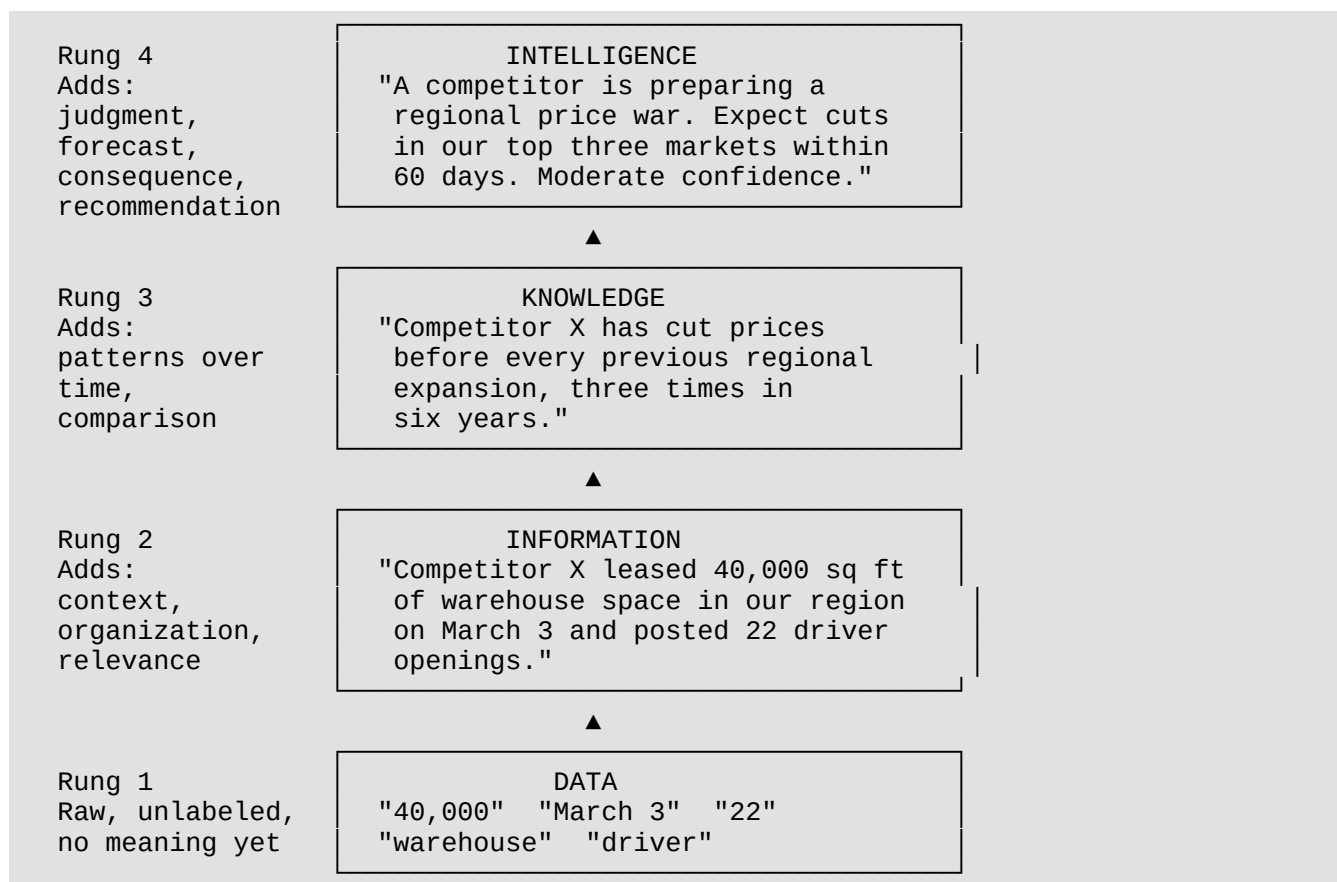
That definition has five working parts, and all five have to be present:

1. **Purpose.** Somebody asked a question. Intelligence answers a question that a decision maker actually has.
2. **Evaluation.** Somebody judged whether the underlying reporting can be trusted.
3. **Analysis.** Somebody added context, comparison, and meaning.
4. **Timeliness.** It arrived before the decision was made, not after.
5. **A judgment.** It says what is probably happening and what it probably means, not just what was observed.

Remove any one of those and you have information again. A brilliant analysis delivered the day after the decision is not intelligence, it is history. A perfectly timed report with no judgment in it is not intelligence, it is a status update.

1.1.2 The Ladder: Data, Information, Knowledge, Intelligence

Intelligence professionals often describe a progression. It helps to visualize it as a ladder where each rung adds something the rung below it lacked.



Read that ladder from the bottom up. At Rung 1 you have numbers and words with no meaning. At Rung 2 those numbers have been attached to entities, dates, and locations, so a human can understand what they describe. At Rung 3 the information has been compared against history and other cases, so patterns appear. At Rung 4 an analyst has said what it means, what happens next, and how sure they are.

Here is the part students most often get wrong: **moving up the ladder is not automatic and it is not free**. Each step takes work, judgment, and a person who is willing to be accountable for a conclusion. Collecting more data does not produce intelligence. It usually just produces a larger pile of data.

1.1.3 The Volume Trap

A common assumption is that more collection produces better intelligence. It does not. Beyond a certain point, additional collection makes analysis harder, because the analyst now has to separate signal from a much larger volume of noise.

Consider a small city police department that installs 200 new cameras. Collection capacity goes up enormously. Analytic capacity does not change at all. The department now records 4,800 camera hours per day and still has two analysts. Unless something changes in processing and analysis, the new cameras produce very little usable intelligence. They mostly produce storage costs and a false sense of coverage.

This is why the intelligence lifecycle, covered in Section 1.2, begins with a requirement rather than

with collection. Collection without a requirement is a hobby.

1.1.4 Reliability and Credibility Are Two Different Questions

Before information can become intelligence, it has to be evaluated. Evaluation has two independent parts that beginners tend to collapse into one.

Source reliability asks: how much do we trust the person, sensor, or organization providing this? A source that has been accurate 30 times in a row is highly reliable. A brand new anonymous tip is unrated.

Information credibility asks: how plausible is this specific report, on its own merits, given everything else we know?

These are separate because a reliable source can be wrong, and an unreliable source can be right. A veteran informant can pass along a rumor they misheard. A first-time anonymous caller can describe something entirely accurate.

Many organizations use a lettered and numbered scheme, sometimes called the Admiralty system or a source evaluation matrix. A simplified version:

Reliability of source	Meaning
A	Completely reliable, long track record, no known failures
B	Usually reliable, minor history of error
C	Fairly reliable, mixed record
D	Not usually reliable, frequent errors
E	Unreliable, history of fabrication or manipulation
F	Cannot be judged, no basis for evaluation

Credibility of information	Meaning
1	Confirmed by other independent sources
2	Probably true, consistent with other reporting
3	Possibly true, plausible but uncorroborated
4	Doubtful, conflicts with other reporting
5	Improbable, contradicted by reliable reporting
6	Cannot be judged

A report rated **B2** is a usually reliable source passing along something that fits the rest of the picture. A report rated **F3** is an unknown source passing along something plausible but unconfirmed. Both can appear in the same product, but the analyst has to weight them differently and the reader deserves to know which is which.

1.1.5 Worked Examples

Study these carefully. You will be asked to do exactly this classification in Lab 1.

Example A: Cybersecurity

Rung	Content
Data	198.51.100.24, 4,412, 03:14, fail
Information	Between 03:14 and 05:02, IP address 198.51.100.24 generated 4,412 failed authentication attempts against the corporate VPN across 310 distinct usernames.
Knowledge	This pattern (many usernames, few passwords each) matches password spraying, and the same address range was reported by a threat sharing group last week as associated with a known access broker.
Intelligence	An external access broker is conducting a password spraying campaign against our VPN. Because account lockout is not enabled on the VPN and 41 accounts still use passwords from a known breach list, we assess with high confidence that at least one account will be compromised within 72 hours unless lockout is enabled and those 41 passwords are reset. Recommend both actions before end of day.

Notice what the intelligence statement adds: an actor, an assessment of intent, a forecast with a time frame, a stated confidence level, the specific vulnerability that makes the forecast credible, and a recommendation the reader can approve in one sentence.

Example B: Retail business

Rung	Content
Data	-3%, Store 14, Tuesday, 1,204
Information	Store 14 foot traffic fell 3% week over week, with 1,204 visitors on Tuesday.
Knowledge	The decline began the week a competitor opened two blocks away, and the same pattern appeared at Stores 6 and 9 after similar openings, with traffic stabilizing at about 12% below baseline after nine weeks.
Intelligence	Store 14 is in the early phase of the same competitive displacement seen at Stores 6 and 9. We assess with moderate confidence that traffic

Rung	Content
	will settle 10% to 14% below baseline by late Q3 without intervention. The two prior locations recovered roughly half the loss through a targeted loyalty campaign in weeks 3 through 6. That window for Store 14 closes in 11 days.

Example C: Emergency management

Rung	Content
Data	18 mph, 9%, 41 C, ridge
Information	Sustained winds of 18 mph from the northeast, relative humidity 9%, temperature 41 C, along the ridge northeast of the county.
Knowledge	These are the same conditions present during the two most destructive fires in county history, both of which crossed the ridge within six hours of ignition.
Intelligence	Conditions northeast of the county are at the historical threshold where ignition becomes a rapid ridge-crossing event. If a fire starts today, we assess with high confidence that the two subdivisions on the downslope side would have less than four hours of warning. The single access road for those subdivisions cannot clear 1,800 vehicles in four hours. Recommend pre-staging engines and issuing a voluntary early evacuation advisory now, not at ignition.

1.1.6 Why This Distinction Matters Practically

Three consequences follow from taking the information versus intelligence distinction seriously.

First, it changes what you ask for. An untrained requester asks for "everything you have on Company X." A trained requester asks "is Company X going to enter our market this year, and what would be the earliest visible sign?" The second question can be answered. The first cannot.

Second, it changes who is accountable. Information carries no accountability. Intelligence does. When an analyst writes "we assess with high confidence," they are putting their professional judgment on the record, and they can be wrong in a way that is later reviewable. This accountability is a feature, not a bug. It is what makes intelligence worth paying for.

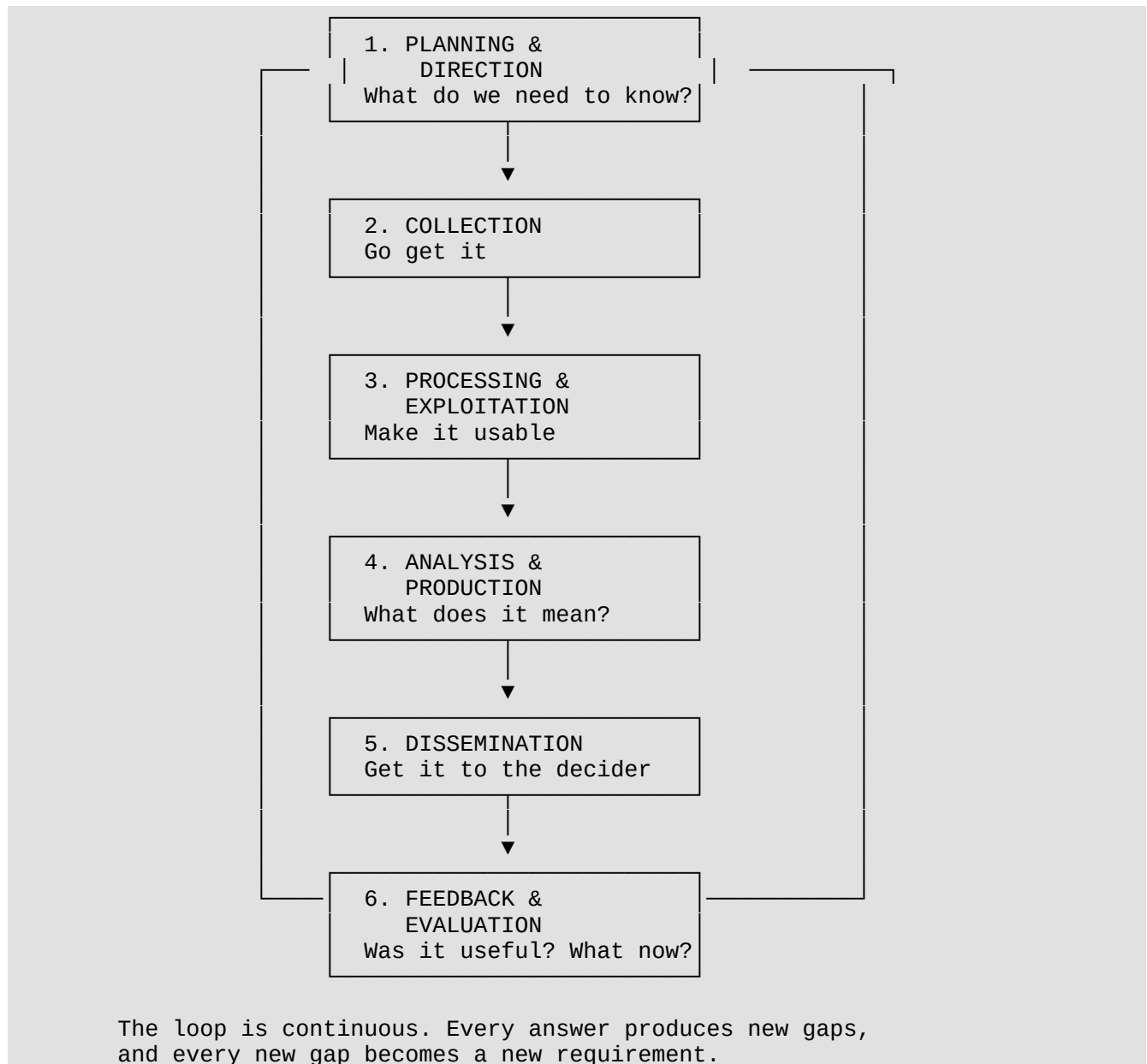
Third, it changes what failure looks like. An organization drowning in dashboards with no analysts is not suffering from an information shortage. It is suffering from an analysis shortage. Those problems have completely different remedies, and buying more collection tools makes the second one worse.

1.2 The Intelligence Lifecycle

1.2.1 Overview

The intelligence lifecycle (also called the intelligence cycle or intelligence process) is the repeating sequence that turns a decision maker's question into a delivered answer, and then turns the answer back into new questions. Different organizations count the phases differently. Some use four, some five, some six, some seven. The names shift, but the logic does not.

This workbook uses six phases plus a feedback loop:



An important correction to the diagram above: in real practice the cycle is **not** a clean circle. Phases run

in parallel, collection continues while analysis is underway, and urgent findings jump straight from collection to dissemination. Practitioners sometimes call the real thing a "tangled web" rather than a cycle. Learn the clean model first, because it gives you the vocabulary. Then expect reality to be messier.

1.2.2 Phase 1: Planning and Direction

This phase converts a vague concern into a specific, answerable requirement.

What happens here:

- A decision maker states a decision they need to make.
- The intelligence team translates that into requirements.
- Requirements are prioritized, because resources are always finite.
- Gaps are identified: what do we already know, and what is missing?
- A collection plan is drafted that assigns each gap to a source or discipline.

Key vocabulary. A **Priority Intelligence Requirement (PIR)** is a question the decision maker has personally endorsed as important. **Essential Elements of Information (EEI)** are the specific sub-questions that must be answered to satisfy a PIR. **Indicators** are the observable things that would tell you the answer is yes or no.

Worked example of decomposition:

Level	Content
Decision	Should we commit \$12M to a distribution center in the Southeast this fiscal year?
PIR	Will our primary competitor enter the Southeast market within 18 months?
EEI 1	Has the competitor acquired or leased regional warehouse capacity?
EEI 2	Is the competitor hiring regional logistics and driver roles?
EEI 3	Has the competitor filed for regional permits or licenses?
EEI 4	Has the competitor's public messaging shifted toward Southeast markets?
Indicators	Commercial lease filings, job postings by metro area, state business registrations, executive travel patterns, trade press mentions, supplier chatter

Why this phase fails. The most common cause of intelligence failure is not bad collection or bad analysis. It is a requirement that was never properly defined. If nobody agreed on the question, no

answer will feel useful. A close second is a requirement that cannot be answered by any available source, which wastes the entire cycle before it starts.

Practitioner's test for a good requirement: it names a decision, it can be answered yes/no/probably, it has a deadline tied to the decision, and it would change what the decision maker does.

1.2.3 Phase 2: Collection

Collection is the deliberate gathering of raw information against the plan produced in Phase 1.

What happens here:

- Sources are tasked according to the collection plan.
- Collection is documented so it can be evaluated later.
- Legal and ethical limits are respected and recorded.
- Gaps that collection cannot fill are reported back to Phase 1.

Types of collection sources:

Source type	Description	Example
Open	Publicly available, no special access	Court filings, job boards, news, satellite imagery for sale
Closed but authorized	Internal to the organization	Payroll data, network logs, sales records
Human	People who report deliberately	Informants, interviews, debriefs, industry contacts
Technical	Sensors and instruments	Cameras, network taps, seismic sensors, telemetry
Shared	Provided by partners under agreement	Threat sharing groups, fusion centers, trade associations

A hard rule. Collection is where legal and ethical failure happens most often, because it is the phase where you touch the outside world. Reading a competitor's public filings is research. Paying their employee for internal documents is theft of trade secrets. Watching a suspect's public social media is open source collection. Installing a tracker on their car without a warrant is a crime. The analytic value of the information does not change the legality of how you got it, and illegally collected information usually destroys the case or contract it was meant to support.

Why this phase fails. Over-collection, which buries analysis. Under-documentation, which makes evaluation impossible later. Single-source dependency, which leaves the whole assessment vulnerable to one liar. And collection that quietly drifts away from the requirement because a source happens to be productive on some other topic.

1.2.4 Phase 3: Processing and Exploitation

Raw collection is rarely usable as collected. Processing makes it usable.

What happens here:

- Translation of foreign language material.
- Decryption or decoding, where lawfully authorized.
- Transcription of audio and video.
- Normalization: putting timestamps, names, and locations into consistent formats.
- Deduplication: recognizing that six reports describe the same event.
- Entity resolution: recognizing that "R. Smith," "Bob Smith," and "Robert A. Smith" are one person.
- Indexing and storage so the material can be retrieved later.

Why this phase is underrated. Processing is invisible when it works and catastrophic when it fails. Two examples of the failure mode:

Time zones. A three-source timeline that mixes UTC, local time, and a vendor's server time will produce a false sequence of events. Analysts will then explain a causal chain that never happened.

Entity resolution. If your system treats one person as three people, you will under-count their activity and miss the pattern entirely. If it merges three people into one, you will invent a conspiracy.

Example. A retail chain collects sales data from 40 stores. Twelve report in local currency, twenty-eight in USD. Nine use a fiscal week starting Sunday, thirty-one use Monday. Until processing normalizes currency and week boundaries, any regional comparison is meaningless. The data is present. The information is not, because it is not yet comparable.

1.2.5 Phase 4: Analysis and Production

This is where information becomes intelligence.

What happens here:

- Evaluation of source reliability and information credibility.
- Integration of multiple sources into a single picture.
- Comparison against baselines and history.
- Generation and testing of competing explanations.
- Assignment of a judgment and a confidence level.
- Writing the product in a form the reader can use.

Structured analytic techniques. Analysts use named methods to reduce the influence of their own biases. A few you should recognize:

Technique	What it does	When to use it
Analysis of Competing Hypotheses (ACH)	Lists all plausible explanations, then scores evidence against each, seeking to disprove rather than confirm	Whenever one explanation feels obvious
Key Assumptions Check	Writes down every assumption the assessment rests on, then asks what happens if each is wrong	Before any high-stakes judgment
Devil's Advocacy	Assigns someone to argue the opposite case in good faith	When a team has converged too quickly
Indicators and Warnings	Defines in advance what would signal a change, then watches for it	For ongoing monitoring requirements
Red Teaming	Models the adversary's decisions from the adversary's perspective and incentives	When forecasting an opponent's behavior
Timeline / pattern analysis	Sequences events to expose cause, opportunity, and anomaly	Criminal, incident, and campaign analysis

Confidence levels. A judgment without a confidence level is unusable, because the reader cannot tell how much weight to put on it. A common scheme:

Confidence	Basis
High	Multiple independent reliable sources agree; few assumptions; strong track record for this type of judgment
Moderate	Credible sourcing with some gaps; plausible alternative explanations exist
Low	Fragmentary or single-source reporting; heavy reliance on assumption; significant alternatives remain open

Confidence describes how solid the *basis* for the judgment is. It is not the same thing as probability. "We assess with high confidence that an attack is unlikely" is a perfectly coherent statement: the analyst is very sure the odds are low.

Estimative language. Precision in probability words matters. Different agencies publish slightly different scales, but the discipline is universal: pick a scale, publish it, and use it consistently.

Phrase	Approximate probability
Almost certainly / nearly certain	95% to 99%

Phrase	Approximate probability
Very likely / highly probable	80% to 95%
Likely / probable	55% to 80%
Roughly even chance	45% to 55%
Unlikely / improbable	20% to 45%
Very unlikely	5% to 20%
Almost certainly not	1% to 5%

Why this phase fails. Confirmation bias, where the analyst notices only what fits the initial theory. Mirror imaging, where the analyst assumes an adversary values what the analyst values. Anchoring on the first report. Layering, where an assumption from an earlier product is treated as established fact in a later one. And politicization, where the judgment is adjusted to match what the reader wants to hear, which is the most damaging failure of all because it destroys the value of every future product.

1.2.6 Phase 5: Dissemination

Intelligence that never reaches the decision maker in a usable form is wasted work.

What happens here:

- Selecting the right format for the audience.
- Selecting the right channel and timing.
- Applying handling and classification controls.
- Confirming receipt and understanding.

Format is not cosmetic. A CEO with six minutes between meetings and a SOC analyst on a night shift need entirely different documents describing the same event.

Product type	Length	Audience	Purpose
Flash / spot report	1 paragraph	Anyone, immediately	Something just happened, act now
Executive brief	1 page	Senior leadership	Decision support, so-what up front
Situation report (SITREP)	1 to 3 pages	Operations staff	Current state and changes since last report
Analytic assessment	5 to 30 pages	Planners, strategists	Deep judgment on a complex question
Indicator watchlist	Table	Watch officers, SOC	What to look for, what to do if seen
Target or entity profile	2 to 10 pages	Investigators, sales, security	Everything known about one subject

The BLUF rule. Write **B**ottom **L**ine **U**p **F**ront. The judgment goes in the first sentence. The evidence follows. Most readers stop after the first paragraph, and that is fine as long as the first paragraph carries the answer. Academic structure, where the conclusion appears at the end after a careful build-up, is exactly wrong for intelligence writing.

Need to know and handling. Dissemination also decides who does *not* receive the product. Over-sharing can burn a source, tip off a subject, or create legal exposure. Under-sharing produces the failure pattern found in nearly every post-incident review ever written: somebody knew, and the person who needed to know did not.

Why this phase fails. Wrong format, wrong length, wrong channel, too late. Also burying the judgment on page nine, and writing in jargon the reader does not speak.

1.2.7 Phase 6: Feedback and Evaluation

The loop closes when the consumer tells the producer whether it helped.

What happens here:

- The decision maker responds: was this useful, did it answer the question, what is still missing?
- The team assesses accuracy after the fact. Did what we forecast actually happen?
- New requirements are generated from the gaps.
- Collection and analysis methods are adjusted.

Why this phase fails. It is skipped. Feedback is the most frequently omitted phase in both government and corporate settings, because everyone is already working the next crisis. The cost is invisible and compounding: without feedback, an intelligence function cannot tell the difference between products that get read and products that get filed, and it will keep producing both.

A practical feedback instrument. Four questions, asked one week after delivery:

1. Did you read it? If not, why not?
2. Did it answer the question you actually had?
3. Did it change anything you did?
4. What question do you have now?

Question 3 is the one that matters. An intelligence product that changed nothing was either wrong, late, obvious, or aimed at the wrong person.

1.2.8 The Lifecycle Under Time Pressure

In a fast-moving incident, the six phases compress but none disappear. Compare a 90-day strategic study with a 90-minute incident response:

Phase	90-day strategic study	90-minute incident
Planning	Weeks of requirement negotiation	60 seconds: "Are we being actively exploited right now?"
Collection	Broad, multi-source, deliberate	Whatever telemetry is already flowing
Processing	Careful normalization and archiving	Fast triage, accept messiness, note it
Analysis	Formal ACH, peer review, coordination	Rapid hypothesis, single reviewer, explicit uncertainty
Dissemination	Formal 20-page assessment	Two sentences in the incident channel
Feedback	Post-project review	Continuous, in the same channel

The lesson is not that the model breaks under pressure. It is that the model tells you what you are trading away when you compress it, so you can say so out loud. "Low confidence, single source, no corroboration yet" is a professional statement. Silence about those gaps is not.

1.3 Intelligence Disciplines

1.3.1 What a Discipline Is

An **intelligence discipline** is a category of collection defined by *where the information comes from* and *how it is gathered*. Disciplines are not topics. There is no "terrorism discipline" or "fraud discipline." There are human sources, signals, images, public material, and so on, and any of them can be pointed at terrorism or fraud.

The community convention is to name disciplines with the suffix "INT," short for intelligence. The core disciplines below originated in national security work, but every one of them has a private sector analogue, which this section makes explicit because that is where most students will actually work.

1.3.2 HUMINT: Human Intelligence

What it is. Information collected from people, through deliberate interaction. Interviews, debriefings, informants, elicitation in conversation, and formal source relationships.

Government form. Case officers recruiting and running sources; military interrogation and debriefing; diplomatic reporting.

Private sector form. Sales conversations at trade shows, exit interviews, customer advisory boards, supplier relationships, industry contacts, structured expert interviews, and information gathered by security staff talking to people on site.

Strengths. Only HUMINT reliably gets at intent, motive, plans not yet acted on, and internal disagreement. A satellite can show you that a factory was built. Only a person can tell you the board

fought about it and the CFO is planning to resign.

Weaknesses. People lie, misremember, exaggerate their access, and tell you what they think you want to hear. Sources can be doubled or fed deliberately false material. HUMINT is slow to develop and carries the highest ethical and legal risk of any discipline.

Ethical line. Asking a competitor's employee about publicly announced products at a conference is normal business conversation. Misrepresenting your identity to obtain confidential information, or inducing someone to violate a nondisclosure agreement, is at minimum a serious ethics violation and often a crime under trade secret law.

1.3.3 OSINT: Open Source Intelligence

What it is. Information collected from publicly and lawfully available sources. This is the largest discipline by volume and the one most accessible to students.

Sources include: news media, academic publications, patent filings, court records, corporate registries and annual reports, regulatory filings, job postings, professional networking profiles, social media, conference presentations, government statistics, commercial satellite imagery, shipping and flight tracking, domain registration records, and code repositories.

Strengths. Cheap, fast, legal, shareable without clearance restrictions, and often the majority of what any assessment needs. Many practitioners estimate that most of the useful content in a typical assessment is open source.

Weaknesses. Enormous volume relative to value. Vulnerable to deliberate deception, because anyone can publish. Vulnerable to circular reporting, where three "independent" sources all trace back to a single unverified original post. Requires real skill to separate credible material from confident nonsense.

A discipline within the discipline. Two OSINT sub-practices matter enough to name. **SOCMINT** (social media intelligence) covers collection from social platforms and carries its own privacy considerations. **GEOINT-from-open-sources** uses publicly purchasable imagery and geolocation of photographs, and has become powerful enough that journalists routinely use it to verify claims about conflicts and disasters.

1.3.4 SIGINT: Signals Intelligence

What it is. Information derived from intercepted signals. Traditionally divided into:

- **COMINT** (communications intelligence): the content and metadata of communications between people.
- **ELINT** (electronic intelligence): emissions from radars and other non-communications electronic systems.
- **FISINT** (foreign instrumentation signals intelligence): telemetry from missiles, aircraft, and space systems.

Government form. National-level interception of foreign communications and emissions, tightly

regulated by law and oversight.

Private sector form. Highly constrained and often illegal. Legitimate corporate analogues are limited to signals on infrastructure you own or are authorized to monitor: your own network traffic, your own wireless environment surveyed for rogue access points, your own telephony metadata.

Strengths. Volume, timeliness, and the ability to observe activity the subject believes is private.

Weaknesses. Encryption blocks content. Metadata alone can be badly misread. And this is the most legally dangerous discipline for private actors. Intercepting communications you are not party to and not authorized to monitor is a criminal act in most jurisdictions.

Student caution. Nothing in this chapter authorizes signals collection. If a lab or capstone scenario involves communications interception, your role is to *identify* that SIGINT would be relevant to a government actor, not to attempt it.

1.3.5 GEOINT and IMINT: Geospatial and Imagery Intelligence

What it is. **IMINT** is intelligence from images: satellite, aerial, drone, and ground photography. **GEOINT** is the broader discipline that combines imagery with maps, elevation data, and other geographically referenced information to describe activity on the earth's surface.

Government form. National reconnaissance satellites, aerial reconnaissance, mapping agencies, battle damage assessment.

Private sector form. Commercial satellite subscriptions used to count cars in retail parking lots, monitor construction progress at a competitor's site, verify supplier facilities, assess flood or fire exposure for insurance, and plan physical security. Drone imagery for facility inspection. Geographic analysis of crime, sales, or risk.

Strengths. Hard physical evidence. Imagery is difficult to fake convincingly at scale and is persuasive to skeptical decision makers. Change detection over time is extremely powerful: the same frame in March and June tells a story neither frame tells alone.

Weaknesses. Shows what, not why. Weather, cloud cover, and orbital timing limit availability. Interpretation requires training, and confident misinterpretation of imagery has caused real historical disasters.

Classic use. During the Cuban Missile Crisis in 1962, aerial photography did not merely suggest that missiles were present. It provided physical proof specific enough to be presented publicly, which changed the diplomatic situation entirely. That is the characteristic contribution of imagery: it converts an argument into an exhibit.

1.3.6 MASINT: Measurement and Signature Intelligence

What it is. Intelligence from the physical characteristics and emissions of a target that are not signals or images: radiation, acoustic signatures, seismic vibration, chemical traces, magnetic and infrared signatures.

Government form. Detecting nuclear tests through seismic and radionuclide monitoring, identifying

submarines by acoustic signature, detecting chemical agent production.

Private sector form. Environmental monitoring, industrial vibration analysis for predictive maintenance, acoustic gunshot detection systems used by cities, radiation portal monitors at ports.

Strengths. Extremely difficult to fake, because it measures physics rather than assertions.

Weaknesses. Narrow applicability, expensive sensors, requires deep technical specialization to interpret.

1.3.7 TECHINT: Technical Intelligence

What it is. Intelligence derived from examining an adversary's equipment, weapons, or products. Reverse engineering, in short.

Government form. Recovering and analyzing captured equipment to understand capability and develop countermeasures.

Private sector form. Teardown analysis of competitor products, which is standard, legal, and widespread in electronics and automotive industries. Buy the competitor's product at retail, disassemble it, cost out the bill of materials, and infer their margin and supply chain.

Strengths. Ground truth about capability. Marketing claims are opinions; a teardown is a fact.

Weaknesses. Requires physical access and specialized labs. Tells you what exists now, not what is coming next.

1.3.8 FININT: Financial Intelligence

What it is. Intelligence derived from financial transactions, records, and flows.

Government form. Suspicious activity reporting, sanctions enforcement, money laundering investigation, terrorist financing analysis.

Private sector form. Anti-money-laundering programs in banks, fraud analytics, credit and counterparty risk assessment, due diligence on acquisition targets and third-party vendors.

Strengths. Money is hard to hide and leaves durable records. Financial flows reveal relationships that participants actively try to conceal.

Weaknesses. Access is legally restricted and rightly so. Cash, informal transfer systems, and some cryptocurrency usage reduce visibility.

1.3.9 CYBINT and CTI: Cyber and Cyber Threat Intelligence

What it is. Intelligence about threats in and through the digital domain: who is attacking, with what tools, using what infrastructure, against which targets, and why. Often labeled **CTI** (cyber threat intelligence) in industry.

CTI is conventionally split into three tiers, which map neatly onto the strategic, operational, and tactical distinction covered in the next section:

Tier	Content	Consumer	Shelf life
Strategic CTI	Which threat actors care about our sector and why; risk trends; regulatory direction	Board, CISO	Months to years
Operational CTI	A specific campaign: who, how, what infrastructure, what stage	SOC leadership, incident response, threat hunters	Weeks to months
Tactical CTI	Indicators of compromise: hashes, IP addresses, domains, detection rules	Detection engineers, automated tooling	Hours to weeks

Strengths. Fast, highly shareable, well supported by mature community sharing arrangements, and directly actionable.

Weaknesses. The industry has an unfortunate habit of confusing indicator feeds with intelligence. A list of 40,000 IP addresses is data. It becomes intelligence when someone tells you which of those addresses matter to your environment, what an appearance would mean, and what to do about it.

1.3.10 Disciplines Work Together, Not Alone

Single-discipline assessments are fragile. **All-source analysis** is the practice of fusing multiple disciplines so that each covers the others' weaknesses.

Consider one question: *is the competitor about to open a regional distribution hub?*

OSINT → Commercial lease filed, 40,000 sq ft, March 3
22 driver job postings in that metro
Trade press quote from a regional VP

GEOINT → Satellite imagery shows dock construction
and new trailer parking, April to June

TECHINT → Teardown of their new product shows a
packaging change suited to regional
distribution rather than direct ship

HUMINT → A supplier mentions being asked to quote
delivery to a new address in that metro

ALL-SOURCE ASSESSMENT
Competitor will open a Southeast hub,
operational Q4. High confidence:
four independent disciplines agree, and
the physical construction cannot be a
feint at this cost.

Notice why confidence is high. It is not because there is a lot of reporting. It is because the reporting comes from sources that cannot easily be manipulated together. Faking a lease filing, a construction site, a packaging change, and a supplier conversation simultaneously would cost more than the hub.

1.4 Strategic Intelligence

1.4.1 Definition and Purpose

Strategic intelligence supports decisions about direction, posture, and resource commitment over long time horizons. It answers questions about where the environment is going and what the organization should become in response.

Its consumer is the top of the organization: heads of state, cabinet officials, boards of directors, chief executives. Its time horizon runs from roughly one year to a decade or more. Its output is judgment about trends, capabilities, intentions, and risks, not about individual events.

1.4.2 Characteristic Questions

- What are the long-term intentions and capabilities of this state, competitor, or threat actor?
- Which structural forces (demographic, technological, regulatory, economic) will reshape our operating environment over the next five years?
- If we commit to this course, what is the range of plausible futures, and which are survivable?
- What are our own long-term vulnerabilities that an adversary would exploit?
- Where should we invest capability that takes years to build?

1.4.3 Distinctive Features

It is estimative. Strategic intelligence deals with things that have not happened yet and may never happen. It trades precision for scope. Anyone demanding certainty at a five-year horizon has misunderstood the product.

It uses scenarios, not single forecasts. Rather than one prediction, good strategic work presents several plausible futures with the indicators that would tell you which one is developing. This is more useful, because it lets leadership pre-decide responses instead of reacting.

It is expensive and slow. A serious strategic assessment takes months, involves many analysts, and requires coordination across disagreeing experts.

It is often unwelcome. Strategic intelligence regularly tells leadership that its existing plan rests on an assumption that is no longer true. Handling that gracefully is a professional skill in itself.

1.4.4 Examples Across Sectors

Sector	Strategic intelligence question
National security	Will this state's military modernization change the regional balance by 2035?
Corporate	Will this technology make our core product category obsolete within a decade?
Cybersecurity	Which categories of threat actor will be motivated to target our sector as we expand into new markets?
Public health	What is our exposure to a pathogen with these transmission characteristics, and what capability should we build now?
Emergency management	How will changing climate and development patterns alter our wildfire and flood exposure over 20 years?
Law enforcement	Which criminal enterprise structures are displacing older ones in our region, and what does that mean for staffing and training?

1.4.5 The Blockbuster Illustration

The clearest way to see strategic intelligence failure is a case where the information was available and the judgment was not made.

Through the early 2000s, the home video rental industry had access to abundant information: broadband adoption rates were climbing, a mail-order rival was growing its subscriber base, and customer complaints about late fees were well documented. The information was public. What was missing was a strategic judgment that connected those trends into a statement of the form "our revenue model depends on a customer behavior that will not survive the next five years of bandwidth growth."

The lesson is not that anyone was foolish. It is that strategic intelligence is the work of drawing a conclusion that is uncomfortable and acting on it while it is still early enough to matter. Late fees were a highly profitable revenue line. A strategic assessment that recommended destroying a profitable line to protect a future business would have been a hard sell in any organization. That is precisely why it needed to be written down, with a confidence level attached.

1.5 Tactical Intelligence

1.5.1 Definition and Purpose

Tactical intelligence supports immediate action. It tells the person doing the work what to do right now, or in the next few hours or days.

Its consumer is the operator: the patrol officer, the SOC analyst, the incident commander at the scene, the sales representative walking into a meeting, the field technician. Its time horizon runs from seconds to days. Its output is specific, concrete, and directly actionable.

1.5.2 Characteristic Questions

- What is happening right now at this location?
- Which door do we enter, and who is inside?
- What is this IP address, and should I block it?
- Who am I about to meet, what do they care about, and what did they last complain about?
- Where is the fire line right now and which street do we evacuate first?

1.5.3 Distinctive Features

Perishability. Tactical intelligence expires fast. A suspect's location is useful for minutes. A malicious IP address may be abandoned by the attacker within days. A product built to be accurate for six months is not tactical intelligence.

Specificity. Tactical products name things: addresses, hashes, license plates, times, names, coordinates. Generalities are useless to an operator.

Brevity. The consumer is busy, often physically occupied, and possibly in danger. A tactical product is a paragraph, a table row, or a radio call.

Low tolerance for hedging. Tactical consumers need a call. Uncertainty must still be stated, but it should be stated in a form that supports action: "assume armed until confirmed otherwise," not "the subject's armament status is presently indeterminate."

1.5.4 Examples Across Sectors

Sector	Tactical intelligence product
Law enforcement	Pre-raid briefing: floor plan, occupants, weapons history, dog, exits
Cybersecurity	Blocklist of 14 domains associated with an active campaign, plus a detection rule
Military	Enemy positions on the far side of a specific ridge, updated hourly
Sales	One-page account brief before a customer call, including the open support ticket
Emergency management	Current fire perimeter, wind shift forecast for the next two hours, road closures
Retail loss prevention	Description and pattern of an organized theft crew currently working the district

1.5.5 Why Tactical Intelligence Fails

The dominant failure is not inaccuracy. It is arriving late, or arriving in the wrong format. A perfectly accurate warning delivered by email to an officer who is already inside the building has failed completely. Tactical intelligence lives or dies on delivery mechanics: the right channel, pushed rather than pulled, in the form the consumer can absorb while doing something else.

The second failure is what practitioners call the indicator treadmill. Because tactical products are perishable, organizations end up producing an endless stream of them and mistake the stream for a program. Without strategic and operational intelligence above it, tactical work becomes an infinite treadmill of blocking individual addresses while the underlying campaign continues untouched.

1.6 Operational Intelligence

1.6.1 Definition and Purpose

Operational intelligence occupies the middle. It supports the planning and management of campaigns, cases, investigations, and programs: bodies of activity larger than a single action but smaller than the whole enterprise strategy.

Its consumer is the middle of the organization: the campaign planner, the case supervisor, the incident response lead, the regional manager, the threat hunting team lead. Its time horizon runs from weeks to roughly a year. Its output describes how an adversary or a market operates, so that a coordinated series of actions can be planned against it.

1.6.2 Characteristic Questions

- How does this criminal network actually function: who supplies, who moves product, who launders, where are the choke points?
- What is this threat actor's full playbook across the intrusion, and where in our environment could we detect them earliest?
- What sequence of actions over the next quarter would disrupt this problem rather than nibble at it?
- How should we allocate our people across this case load for the next six months?
- What resources will this campaign require, and what will the adversary do in response?

1.6.3 Distinctive Features

It is about structure and process, not events. Tactical intelligence says "this address is malicious." Operational intelligence says "this actor gains access through supplier portals, moves laterally within 40 minutes, and always stages data in cloud storage before exfiltration, which means our earliest reliable detection point is the supplier portal, not the exfiltration."

It is the bridge. Operational intelligence translates strategic direction into tactical tasking, and

aggregates tactical observations into patterns leadership can act on. When an organization has strategic and tactical intelligence but nothing in the middle, the two ends stop talking. Leadership sets direction that never reaches the operators, and the operators generate observations that never reach leadership.

It is where disruption gets designed. Arresting a street-level dealer is tactical. Understanding that one wholesale supplier serves nine neighborhoods, and sequencing a case to remove that supplier, is operational.

1.6.4 Examples Across Sectors

Sector	Operational intelligence product
Law enforcement	Network chart of a trafficking organization with roles, relationships, and enforcement choke points
Cybersecurity	Campaign assessment mapping an actor's full technique chain to our detection coverage, with gaps
Military	Assessment of an adversary's logistics and sustainment across a theater over a season
Business	Analysis of a competitor's go-to-market motion and which channel partners are vulnerable to being flipped
Emergency management	Multi-week resource and mutual aid plan for a season based on projected event frequency
Public health	Transmission network analysis for an outbreak, guiding where to concentrate intervention

1.6.5 The Most Commonly Missing Layer

If you audit real intelligence functions, operational intelligence is the layer most often absent. The reasons are structural. Tactical work is urgent, so it always gets done. Strategic work is visible to executives, so it gets funded. Operational work is neither urgent nor executive-facing, so it gets deferred.

The symptom of a missing operational layer is an organization that is extremely busy and never gets ahead of anything. It responds to the same category of problem repeatedly, with competent tactical execution each time, and never asks the operational question: what is the structure that keeps generating these events, and how do we take a piece of it away?

1.6.6 A Note on Conflicting Definitions

Be aware that the word *operational* is used differently in different communities, and this trips up students constantly.

- In the classic military framing, the three levels are strategic (national or theater objectives), operational (campaigns within a theater), and tactical (individual engagements). This is the

usage this chapter follows.

- In some business and IT contexts, "operational" is used loosely to mean "day to day," which pushes it toward what this chapter calls tactical.
- In cyber threat intelligence, "operational" specifically means campaign-level and actor-level analysis, which matches the classic framing.

When you encounter the term in the field, check which definition is in play before assuming. On an exam, use the definitions in this chapter.

1.7 Competitive Intelligence

1.7.1 Definition

Competitive intelligence (CI) is the lawful and ethical collection, analysis, and dissemination of information about competitors, customers, suppliers, regulators, and market forces, used to support business decisions.

The emphasis on *lawful and ethical* is not decorative. It is the defining boundary of the profession. Competitive intelligence and industrial espionage use overlapping analytic methods and entirely different collection methods, and confusing them ends careers and companies.

1.7.2 What Competitive Intelligence Covers

Students often assume CI means "watching competitors." It is broader than that. A standard framing covers five forces in the operating environment:

Focus area	Typical questions
Competitors	Who are they, what can they do, what will they do next, what are they bad at?
Customers	What is changing in what they need, what would make them switch?
Suppliers	Where are we dependent, who else is bidding for their capacity, what is their financial health?
New entrants	Who could enter this market, what would it cost them, are they moving?
Substitutes and regulation	What could replace our category entirely, what rule changes are coming?

1.7.3 The Ethical and Legal Boundary

This table is worth memorizing. Everything in the left column is normal professional practice. Everything in the right column can result in civil liability, criminal charges, or both.

Legitimate competitive intelligence	Industrial espionage or unfair practice
Reading published annual reports, filings, and patents	Obtaining unpublished internal financials through a paid insider
Buying and disassembling a competitor's retail product	Breaking into a facility, or accessing systems without authorization
Attending public conference talks and taking notes	Recording a closed session you were not authorized to attend
Analyzing job postings to infer strategy	Posing as a recruiter to extract confidential details from an employee
Interviewing a former employee about publicly known matters	Inducing a former employee to disclose material covered by an NDA
Talking to shared suppliers about general market conditions	Paying a supplier to breach confidentiality about a competitor
Counting cars in a public parking lot, or buying commercial satellite imagery	Trespassing, placing hidden cameras, or installing trackers
Identifying yourself accurately and asking questions	Misrepresenting who you are or who you work for

Two rules that resolve most gray areas.

The **identity rule**: if you would have to lie about who you are or who you work for to obtain the information, do not obtain it.

The **publication rule**: if the information's owner took reasonable measures to keep it secret, and you are trying to get around those measures, you are outside the profession. This is not merely an ethical formulation. It closely tracks how trade secret law defines protectable information.

Professional bodies in the field publish codes of conduct that say substantially this. Many corporate CI functions require analysts to sign one, and a well-run function keeps a written record of where each significant finding came from, precisely so that it can prove the finding was lawfully obtained if challenged.

1.7.4 Where Competitive Intelligence Sits Relative to the Levels

Competitive intelligence is not a fourth level alongside strategic, tactical, and operational. It is a *domain* that has all three levels inside it:

Level	Competitive intelligence example
Strategic	Will this category be commoditized within five years, and who is positioned to win if it is?
Operational	How does this competitor's channel partner program actually work, and which three partners are unhappy enough to move?

Level	Competitive intelligence example
Tactical	This competitor is discounting 15% in two of our accounts this month; here is the specific counter-offer authority the rep needs today

1.7.5 Counterintelligence for Businesses

The mirror image of CI is protecting your own organization from someone else's CI. This is sometimes called defensive competitive intelligence or corporate counterintelligence. Practical measures include:

- Auditing your own public footprint: what do your job postings, patent filings, and conference talks reveal about your roadmap?
- Training employees on elicitation, which is the technique of extracting information through ordinary-seeming conversation.
- Controlling what is visible at trade shows and in facility tours.
- Reviewing supplier and partner contracts for confidentiality obligations that actually bind.
- Recognizing that your own job postings are frequently the single richest open source about your strategy, because they must be specific to be effective.

1.8 Criminal Intelligence

1.8.1 Definition

Criminal intelligence is the product of collecting and analyzing information about criminal activity, offenders, and criminal networks in order to prevent, disrupt, and prosecute crime, and to allocate law enforcement resources effectively.

It differs from ordinary case investigation in orientation. A criminal investigation asks "who committed this specific act, and can we prove it?" Criminal intelligence asks "what is the structure and pattern of this criminal activity, and where would intervention do the most good?"

1.8.2 Intelligence-Led Policing

The organizing philosophy in modern practice is **intelligence-led policing (ILP)**: the idea that enforcement resources should be directed by analysis rather than by reactive response volume alone.

ILP rests on a few empirical findings that have held up well across jurisdictions:

- Crime concentrates heavily in a small number of places, often called hot spots.
- A small number of offenders account for a disproportionate share of offenses.
- A small number of victims account for a disproportionate share of victimization.

If those three concentrations are real, then directing resources at the concentrations rather than

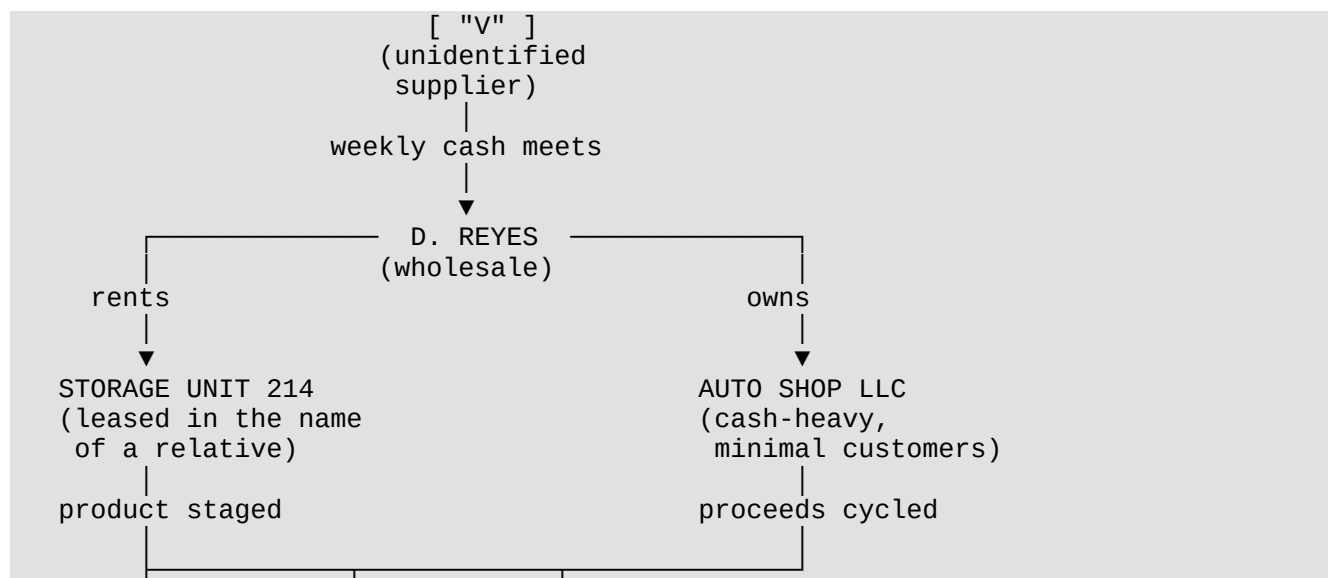
spreading them evenly produces more prevention per officer hour. That is an intelligence problem, because finding the concentrations requires analysis.

1.8.3 Core Analytic Products

Product	What it shows	Level
Crime pattern analysis	Series of related offenses linked by method, timing, or geography	Operational
Hot spot / geographic analysis	Where offenses concentrate, mapped over time	Operational
Link analysis chart	Relationships among people, phones, vehicles, addresses, and businesses	Operational
Network / social network analysis	Structure of an organization: leadership, brokers, choke points	Operational
Target profile	Everything known about one priority offender	Tactical to operational
Threat assessment	Comparative ranking of criminal groups by harm and capability	Strategic
Problem profile	Full analysis of a recurring crime problem and its drivers	Operational to strategic
Bulletin / BOLO	Immediate description and instruction for patrol	Tactical

1.8.4 A Link Analysis Diagram, Described

Link charts are the most recognizable criminal intelligence product. A simplified example, rendered in text:





READING THE CHART:

- Removing a retail node (Okafor, Hall, Price) replaces itself in days.
- Removing Reyes disrupts three retail nodes at once.
- The auto shop is the financial choke point and the strongest evidentiary target, because financial records are durable.
- "V" is the highest-value unknown. Identifying V is the top collection requirement, not an afterthought.

That last point is the difference between a chart and intelligence. A chart displays what is known. Intelligence says which unknown matters most and what would resolve it.

1.8.5 Legal and Civil Liberties Constraints

Criminal intelligence operates under constraints that have no equivalent in business intelligence, and for good reason: the state can deprive people of liberty.

Key principles that appear in most democratic legal systems and in professional standards:

- **Criminal predicate.** Information about individuals should be retained in intelligence systems only when there is a reasonable suspicion of criminal involvement. Collecting and retaining files on people because of their political views, religion, or associations, absent criminal predicate, is both unlawful in many jurisdictions and corrosive to public trust.
- **Purpose limitation and retention limits.** Records should be kept for a defined purpose and purged on a schedule.
- **Audit and accountability.** Who queried what, and why, should be recorded and reviewable.
- **Separation of intelligence from evidence.** Intelligence can guide where to look. It is not itself proof, and using it as proof without independent evidence is a due process problem.
- **Warrant requirements.** Techniques that intrude on a reasonable expectation of privacy generally require judicial authorization.

Predictive policing caution. Analytic systems that forecast where or by whom crime will occur can inherit and amplify bias present in historical arrest data. If a neighborhood was historically over-policed, the arrest record will show more crime there, and a model trained on that record will direct more policing there, generating more arrests. Students should understand this feedback loop as a technical problem with civil rights consequences, not as an abstract complaint. Good practice includes auditing input data, measuring outcomes by community, and keeping a human accountable for every decision the system influences.

1.9 Business Intelligence

1.9.1 Definition and the Naming Confusion

Business intelligence (BI) is the set of technologies, processes, and practices used to collect, integrate, analyze, and present an organization's own data to support management decisions.

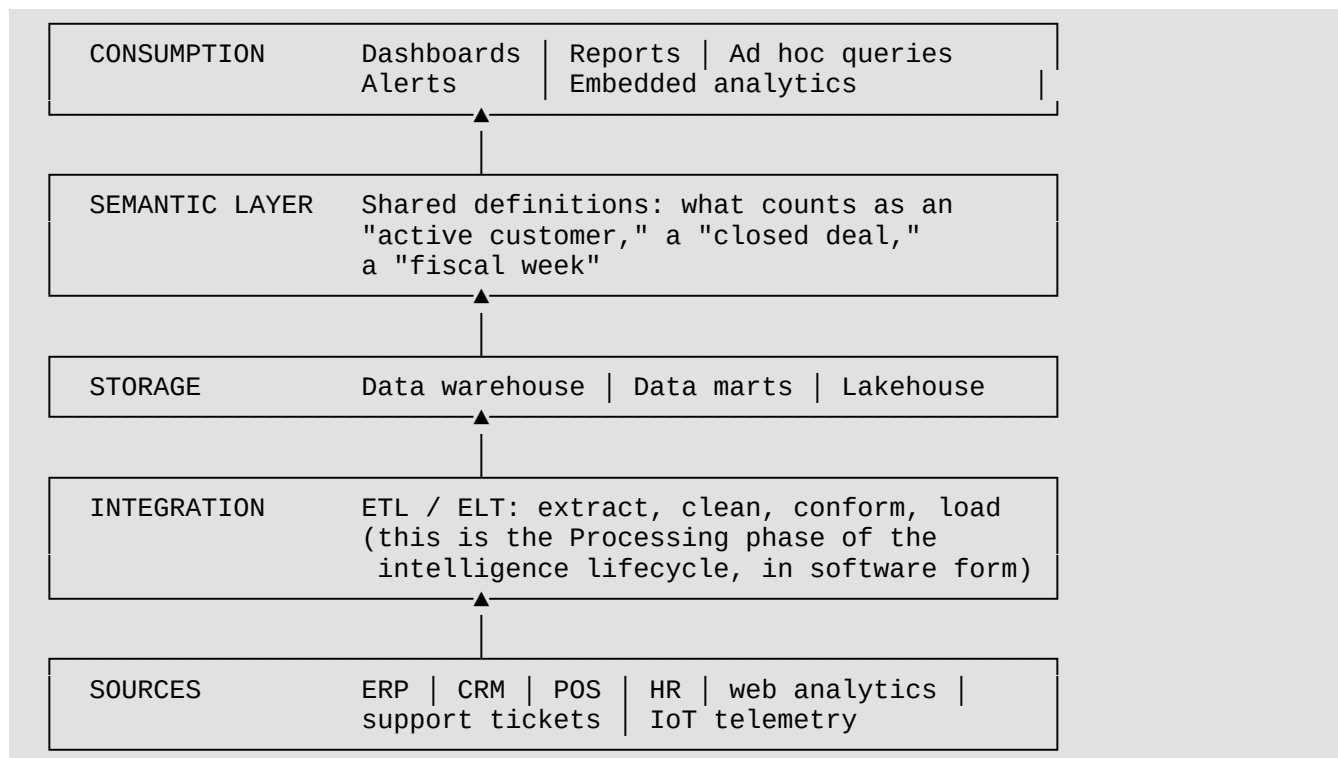
Here is the confusion students must resolve. In the intelligence tradition, "intelligence" means analyzed judgment about an external environment. In the technology industry, "business intelligence" refers primarily to internal reporting and analytics infrastructure: data warehouses, dashboards, and reports.

The two uses of the word are not the same thing, and both are legitimate in their own communities. The practical distinction:

- **Business intelligence** looks *inward* at what the organization did, using data the organization already owns.
- **Competitive intelligence** looks *outward* at what others will do, using external sources.

An organization needs both. BI tells you your Southeast margin fell four points last quarter. CI tells you a competitor is about to enter the Southeast. Neither answers the other's question, and a company with excellent BI and no CI is very well informed about its own past.

1.9.2 The BI Stack, Described as a Diagram



Compare that stack to the intelligence lifecycle and the parallel is exact. Sources are collection. Integration is processing. The semantic layer is a form of evaluation, because it decides what the numbers mean. Consumption is dissemination. The layer that BI stacks most often omit is analysis: the

human judgment that turns a chart into a recommendation. That omission is why so many organizations own hundreds of dashboards and still make decisions on instinct.

1.9.3 Descriptive, Diagnostic, Predictive, Prescriptive

A standard maturity framing, which maps loosely onto the data-to-intelligence ladder from Section 1.1:

Stage	Question answered	Example	Ladder rung
Descriptive	What happened?	Q2 revenue was down 6% in the Southeast	Information
Diagnostic	Why did it happen?	The decline tracks two store openings by a competitor	Knowledge
Predictive	What will happen?	Without action, traffic settles 12% below baseline by Q4	Intelligence
Prescriptive	What should we do?	Run the loyalty campaign in weeks 3 through 6, which recovered half the loss previously	Intelligence with recommendation

Most organizations live at descriptive and think they are further along. The test is simple: can you name a decision that changed last quarter because of a dashboard? If not, the stack is producing information, not intelligence.

1.9.4 Common Failure Modes in BI

- **Dashboard proliferation.** Hundreds of dashboards, no owner, no one certain which number is correct.
- **Metric drift.** Two departments define "active customer" differently, so their reports disagree and both are technically right.
- **Vanity metrics.** Measures that always go up and never inform a decision.
- **No so-what.** A chart with no accompanying judgment leaves interpretation to whoever is loudest in the meeting.
- **Garbage in.** No amount of visualization fixes a source system that records data inconsistently. Processing failures propagate upward invisibly.

1.10 Cross-Cutting Practices You Will Be Graded On

Before the exercises, four practices apply to every product you will write in this chapter.

Dimension	Information	Intelligence
Context	Isolated	Integrated with history, baselines, and other sources
Content	What was observed	What it means and what happens next
Judgment	None	Explicit assessment with confidence level
Uncertainty	Unstated	Stated, with assumptions and gaps named
Audience	Anyone or no one	A specific consumer
Timing	Whenever	Before the decision
Accountability	None	The analyst owns the judgment
Volume	Grows without limit	Deliberately small
Test of quality	Is it accurate?	Did it change a decision for the better?
Failure mode	Missing or wrong	Late, unread, ignored, or hedged into uselessness

2.2 Strategic Versus Operational Versus Tactical Intelligence

Dimension	Strategic	Operational	Tactical
Core question	Where is this going and what should we become?	How does this thing work and how do we disrupt it?	What do I do right now?
Time horizon	1 to 10+ years	Weeks to 1 year	Seconds to days
Consumer	Board, executives, cabinet, chiefs	Campaign planners, case supervisors, IR leads, regional managers	Operators, patrol, SOC analysts, field staff, reps
Focus	Trends, capabilities, intentions, risk posture	Structure, process, networks, campaigns	Events, entities, locations, indicators
Typical product	Multi-page assessment, scenario set, annual threat assessment	Network chart, campaign assessment, problem profile	Bulletin, blocklist, pre-operation brief, spot report
Product length	5 to 50 pages	2 to 20 pages	1 paragraph to 2 pages
Perishability	Low; useful for years	Medium; useful for months	High; may expire in hours
Specificity	Low; deals in categories and ranges	Medium; deals in structures and roles	High; names exact things
Precision expected	Directional	Structural	Exact

Dimension	Strategic	Operational	Tactical
Sources	Broad all-source, heavy on open source and expert judgment	Fused multi-source over time	Whatever is current and fast
Analytic methods	Scenarios, ACH, key assumptions, red teaming	Network analysis, pattern analysis, technique mapping	Triage, matching, correlation
Production time	Weeks to months	Days to weeks	Minutes to hours
Cost per product	High	Medium	Low
Consequence of being wrong	Strategic misallocation discovered years later	A campaign that fails to disrupt anything	An immediate bad action, possibly dangerous
Cybersecurity example	Which actor classes will target our sector as we expand	This actor's full technique chain versus our detection coverage	Block these 14 domains, deploy this rule
Law enforcement example	Which criminal enterprise types are displacing others regionally	Structure of one trafficking network and its choke points	Pre-raid briefing for tonight's warrant
Business example	Will this category be commoditized in five years	How the competitor's partner channel actually works	Competitor is discounting 15% in these two accounts today

2.3 Government Versus Business Intelligence

Dimension	Government / national security	Business / private sector
Ultimate purpose	National security, public safety, policy advantage	Profit, growth, risk reduction, competitive advantage
Primary consumer	Policymakers, commanders, prosecutors	Executives, boards, product and sales leadership
Legal authority	Statutory authority to collect, including intrusive methods with oversight	No special authority; limited to lawful public and owned sources
Available disciplines	Full range including SIGINT, HUMINT with recruited sources, national IMINT	Mostly OSINT, TECHINT via teardown, internal data, commercial imagery, ethical HUMINT
Constraints	Classification, oversight bodies, civil liberties law, warrant requirements	Trade secret law, privacy law, contract and NDA obligations, professional codes

Dimension	Government / national security	Business / private sector
Consequence of overreach	Constitutional violation, oversight action, criminal exposure	Civil liability, criminal charges under trade secret statutes, reputational destruction
Information handling	Formal classification levels and compartments	Confidentiality tiers, need to know, insider trading controls
Time pressure	Varies; strategic products may take a year	Usually compressed; a quarter is a long horizon
Tolerance for uncertainty	High; estimative language is institutionalized	Low; executives often expect a single answer
Analytic workforce	Career analysts, formal tradecraft training, structured methods	Often generalists in strategy, marketing, or security with little formal tradecraft training
Feedback loop	Formal post-mortems, inspector general and legislative review	Rare and informal; often none at all
Sharing	Interagency channels, classification-controlled	Trade associations, sector sharing groups, vendor feeds
Failure visibility	Public inquiries and commissions	Quiet; a lost market is rarely traced to an intelligence gap
What each does better	Depth, tradecraft rigor, hard-target collection	Speed, direct proximity to the decision maker, commercial tooling

2.4 Intelligence Disciplines Compared

Discipline	Collects	Best at	Weakest at	Cost	Legal risk (private sector)	Private sector analogue
OSINT	Publicly available material	Breadth, speed, shareability, context	Deception, circular reporting, volume	Low	Low	Market and competitor research
HUMINT	Information from people	Intent, plans, internal dissent	Deception, bias, slow to build	Medium to high	High	Trade shows, expert interviews, exit interviews
SIGINT	Intercepted signals and emissions	Volume, timeliness, private activity	Encryption, metadata misreading	Very high	Very high; generally unlawful	Monitoring of owned networks only

Discipline	Collects	Best at	Weakest at	Cost	Legal risk (private sector)	Private sector analogue
IMINT / GEOINT	Images and geospatial data	Physical proof, change over time	Explaining why, weather and timing limits	Medium	Low if commercial and public	Commercial satellite, drone survey, geographic analysis
MASINT	Physical signatures and emissions	Hard to spoof, technical ground truth	Very narrow use, deep specialization	High	Low to medium	Environmental and industrial sensing
TECHINT	Adversary or competitor hardware	Capability ground truth	Requires physical access; backward looking	Medium	Low if lawfully purchased	Product teardown and cost analysis
FININT	Transactions and financial records	Hidden relationships, durable records	Restricted access, cash and informal transfer	Medium	High; heavily regulated	AML, fraud analytics, vendor due diligence
CYBINT / CTI	Digital threat activity and infrastructure	Speed, actionability, community sharing	Confusing feeds with analysis, attribution difficulty	Low to medium	Low if defensive and scoped	Threat intelligence program

Part 3: Case Studies

Each case study follows the same structure so you can compare them: the situation, the requirement, the disciplines used, the lifecycle in action, the outcome, and what would likely have happened without intelligence. Two are historical. Three are composites built from documented patterns, and they are labeled as composites so you never cite them as real events.

Case Study 1: Government Decision Making

The Cuban Missile Crisis, 1962 (historical)

The situation. In the autumn of 1962, United States policymakers faced reporting that the Soviet Union might be placing offensive missiles in Cuba. Human source reporting from the island was suggestive but fragmentary, and refugee accounts were plentiful, inconsistent, and impossible to verify

individually. Soviet officials denied any offensive deployment.

The requirement. The decision facing the president was whether to take military action, pursue a blockade, negotiate, or do nothing. The intelligence requirement supporting that decision was narrow and answerable: are offensive nuclear-capable missiles being installed in Cuba, and how close are they to operational readiness?

Notice how well formed that requirement is. It names a decision, it can be answered, and the answer changes what the decision maker does.

Disciplines used.

Discipline	Contribution
HUMINT	Early indications from sources on the island and from refugee reporting; suggested where to look
IMINT	High-altitude aerial photography that produced definitive physical evidence of missile site construction
Additional analysis	Comparison of the photographed layouts against known Soviet deployment patterns, which allowed analysts to identify the systems and estimate readiness timelines

The lifecycle in action.

Phase	What happened
Planning and direction	The question was defined tightly: offensive missiles, yes or no, and readiness timeline
Collection	Reconnaissance flights tasked over specific areas that human reporting had flagged
Processing	Film developed and interpreted by trained photo interpreters, a specialist skill
Analysis	Site layouts matched to known Soviet configurations; readiness estimated; alternative explanations considered and rejected
Dissemination	Briefed directly to the president and the executive committee, with the photographs themselves as exhibits
Feedback	Continued daily collection and reassessment throughout the crisis as the situation evolved

Why the intelligence mattered.

Three things changed because of the analysis rather than because of the raw imagery.

First, **it converted a disputed claim into an established fact.** Denials became untenable once the physical evidence existed and could be shown.

Second, **it produced a timeline.** Knowing roughly how long until the sites became operational defined the size of the decision window. A decision window is one of the most valuable things intelligence can give a leader, because without it every option looks equally urgent.

Third, **it enabled a graduated response.** Because leadership had specific knowledge of what existed and how far along it was, they could select a measured option rather than choosing between doing nothing and doing everything. Intelligence expanded the option set.

The counterfactual. Without the imagery, policymakers would have been choosing among options based on contested human reporting and adversary denials. Either an overreaction based on unverified claims or an underreaction based on accepted denials becomes far more likely. The historical value of this case is that it shows intelligence functioning as intended: a specific question, the right discipline applied to it, delivered in time to shape a decision.

Discussion questions.

1. Which single discipline was decisive, and why could the others not have substituted for it?
 2. The requirement was unusually well specified. Write a badly specified version of the same requirement and explain what would have gone wrong.
 3. Imagery answered "what." Which questions remained that only HUMINT could address?
-

Case Study 2: Business Competition

Meridian Foods and the Regional Price War (composite)

This is a composite case built from common, documented competitive intelligence patterns. It is not a real company.

The situation. Meridian Foods is a regional packaged-goods company with roughly \$400M in revenue across four states. Its largest competitor, a national brand, has entered two of Meridian's four states in the past three years. In both cases, Meridian lost share rapidly in the first two quarters and never fully recovered.

The requirement. The CEO's decision: whether to pre-emptively invest \$18M in a regional brand and trade-spend program in State Three, before any competitor move. The requirement: will the national competitor enter State Three within the next four quarters, and if so, what will their entry sequence look like?

Disciplines used.

Discipline	Collection activity	Finding
OSINT	Commercial real estate filings and permit records	A distribution facility lease signed in the State Three metro

Discipline	Collection activity	Finding
		area
OSINT	Job postings by metro, tracked weekly for nine months	Territory sales manager and merchandiser roles appearing in State Three, a pattern that preceded both prior entries by roughly five months
OSINT	Trade press and investor call transcripts	Executive language shifting toward "underpenetrated adjacent geographies"
GEOINT	Commercial satellite imagery of the leased facility	Dock expansion and trailer staging consistent with an operational hub, not a warehouse held for speculation
HUMINT (ethical)	Conversations with shared brokers and distributors at a trade show, with accurate self-identification	Requests for quotes on State Three delivery lanes
TECHINT	Retail purchase and teardown of competitor packaging	New pack configuration matched to club-store formats dominant in State Three
Internal BI	Meridian's own share and price data from the two prior entry states	A repeatable entry playbook: deep introductory discounting in club and mass channels for two quarters, then price normalization once share was captured

The lifecycle in action.

- **Planning:** the CEO's decision produced one PIR and four EEIs, with named indicators for each.
- **Collection:** an eight-month collection plan across public records, job boards, imagery, and industry conversation, with a written log of where each finding came from.
- **Processing:** job postings normalized by metro and role family so that the signal was comparable to the two prior entries.
- **Analysis:** the analyst ran a competing hypotheses check. Alternative explanations, including a third-party logistics arrangement unrelated to entry and a defensive facility to serve an existing state, were tested and judged less likely because the sales-role hiring pattern only makes sense for a market being entered.
- **Dissemination:** a one-page executive brief with the judgment in the first sentence, a timeline graphic, and a single recommended decision.

- **Feedback:** the CEO asked one follow-up question, which became a new requirement: what is the earliest observable sign that they have moved the entry date forward?

The assessment delivered. Entry into State Three is *likely* (roughly 70%) within three quarters, with **moderate to high confidence**, based on four independent disciplines pointing the same direction and a hiring pattern that matched both prior entries. Expected sequence: club and mass channel first, with introductory discounting in the range of 20% to 25% for approximately two quarters.

How the outcome improved.

Because Meridian had a forecast with a sequence attached, it did three things it could not have done reactively:

1. **Locked in shelf commitments** with two regional grocery chains before the competitor could bid for the same space, using multi-year terms.
2. **Redirected the \$18M** from a broad brand campaign into targeted loyalty and trade programs in the two specific channels the competitor was expected to attack first. Same money, aimed at the right place.
3. **Pre-approved a price response authority** for the field sales team, so that when discounting began, reps could respond within a day rather than waiting weeks for executive approval.

Meridian lost share when the entry came, because entrants with national scale usually take some share. It lost roughly a third of what it had lost in the two prior states, and it retained its shelf position.

The counterfactual. Reactive response means learning about entry from a sales report showing share loss, which is a lagging indicator by definition. By then the shelf space is committed, the competitor's discounting is already live, and the defender is negotiating from weakness.

Note on ethics. Every collection activity in this case was lawful and would survive disclosure. That is the working test for competitive intelligence: could you describe how you got this in a deposition without discomfort?

Discussion questions.

1. Which finding was the strongest single indicator, and why is a sales-role hiring pattern more diagnostic than a lease?
 2. The analyst assigned moderate to high confidence rather than high. Argue for and against that choice.
 3. Identify one collection activity that would have crossed the ethical line, and explain what specifically makes it improper.
-

Case Study 3: Criminal Investigation

Focused Deterrence and Youth Homicide, Boston, 1990s (historical)

The situation. In the early 1990s, Boston experienced a severe spike in youth firearm homicide. The conventional response available to the city was broad enforcement: more patrol, more arrests, more prosecutions distributed across the affected neighborhoods.

The requirement. Rather than asking "how do we make more arrests," the working group of researchers and practitioners asked an analytic question: who is actually driving this violence, through what mechanism, and where would intervention interrupt it?

That reframing is the heart of the case. It converted an enforcement problem into an intelligence problem.

Disciplines and methods used.

Method	Application
Criminal records and incident analysis	Systematic review of homicide cases and the criminal histories of victims and offenders
Practitioner debriefing (a form of HUMINT)	Structured interviews with probation officers, gang unit officers, and street outreach workers who had direct knowledge
Social network analysis	Mapping which groups were in conflict with which, and how disputes propagated between them
Geographic analysis	Locating where violence concentrated

What the analysis found. A very small population of chronically offending young people, concentrated in a modest number of identifiable groups, accounted for a large share of the homicides. The violence was substantially driven by ongoing disputes between groups, propagating in cycles of retaliation, rather than by a diffuse population of unrelated offenders.

That finding is not obvious from arrest volume alone. It required linking people, groups, incidents, and conflicts together, which is operational-level analysis.

The lifecycle in action.

- **Planning:** the requirement was defined as understanding the mechanism of the violence, not counting it.
- **Collection:** existing records plus systematic debriefing of practitioners who held knowledge that was never written down anywhere.
- **Processing:** case files and interview content reduced to a common structure so that people, groups, and conflicts could be linked.
- **Analysis:** network and pattern analysis identifying the small offending population and the

retaliatory dynamic.

- **Dissemination:** findings presented to an interagency working group that included police, prosecutors, probation, clergy, and community organizations, which meant the intelligence reached people who could act through different levers.
- **Feedback:** the strategy was monitored and adjusted, and the analytic loop continued as conditions changed.

The intervention the intelligence enabled. Because analysis identified specific groups rather than general neighborhoods, the response could be a direct communication strategy: tell the identified groups clearly and in advance that further violence by any member would bring concentrated attention from every available legal lever against the whole group, while simultaneously offering services and support to those who wanted out.

This design is only possible with intelligence. You cannot deliver a specific, credible warning to a group you have not identified, and you cannot make the warning credible unless you actually know who is who.

How the outcome improved. Boston recorded a substantial decline in youth homicide following implementation, and the approach, generally known as focused deterrence or the pulling-levers strategy, was studied and replicated in other cities with varying results. Reviews of the broader body of focused deterrence research have generally found meaningful crime reduction effects, while also noting that results depend heavily on implementation quality and sustained analytic support.

The counterfactual. Broad enforcement across affected neighborhoods spreads resources thinly, generates large numbers of low-consequence arrests, imposes costs on the majority of residents who are not involved in violence, and does not interrupt the retaliation cycle that generates the next homicide.

Cautions this case teaches. Two of them, and both matter.

First, **the analysis has to stay current.** Group structures and conflicts change. A network chart from two years ago will direct resources at a structure that no longer exists.

Second, **precision creates responsibility.** Identifying specific individuals for concentrated attention is a significant exercise of state power. It demands rigorous predicate, documented reasoning, auditability, and genuine availability of the service and support side of the strategy. A focused deterrence program with the enforcement half and none of the outreach half is not the strategy that was studied.

Discussion questions.

1. Why was practitioner debriefing indispensable here, and what does that tell you about where organizational knowledge actually lives?
2. Classify this work as strategic, operational, or tactical, and defend the classification.
3. What are the civil liberties safeguards you would require before implementing this in your own jurisdiction?

Case Study 4: Cybersecurity Incident

Cascade Regional Health and a Pre-Ransomware Intrusion (composite)

Composite case built from documented ransomware intrusion patterns. Not a real organization.

The situation. Cascade Regional Health operates four hospitals and 22 clinics. Its security operations center is small: five analysts, one of whom has threat intelligence as a part-time responsibility. In March, the SOC observed an unremarkable alert: a service account authenticating from a workstation it had never used before, at 02:40.

Alone, that alert is information. It had fired 60 times in the previous quarter and had been a false positive every time.

The requirement. The part-time intelligence analyst had, three months earlier, produced a strategic assessment stating that ransomware operators were actively targeting regional healthcare providers, and that operators in this class typically used one of a small number of initial access methods followed by a predictable escalation sequence. From that strategic product came a standing operational requirement: identify activity matching the early stages of that sequence *before* encryption, and define the indicators that would signal it.

That standing requirement is what turned a routine alert into an investigation.

Disciplines used.

Discipline	Contribution
Strategic CTI	Sector targeting assessment that established ransomware operators as a priority threat and named their general playbook
Operational CTI	The specific technique chain: valid account access, discovery activity, credential dumping, lateral movement to backup infrastructure, then encryption
Tactical CTI	Indicators shared through a health sector information sharing group: file hashes, remote access tool names, and command line patterns
Internal telemetry (owned-network monitoring)	Endpoint logs, authentication logs, network flow data
FININT (adjacent)	Finance flagged that the same service account had been used to alter a vendor payment record, which linked the intrusion to an attempted fraud

The lifecycle in action.

Phase	Detail
Planning	Standing requirement: detect pre-encryption stages of the known playbook. Named indicators defined in advance
Collection	Existing endpoint and authentication telemetry, plus a sharing-group indicator feed
Processing	Log timestamps normalized to UTC (a prior incident had produced a wrong timeline because of mixed time zones), and account name variants resolved to single identities
Analysis	The lone alert was correlated against the operational playbook. Three additional matching behaviors were found in existing logs that no one had connected: discovery commands run from the same workstation two days earlier, an unusual authentication to the backup server, and a remote access tool matching a shared indicator
Dissemination	A four-sentence flash report to the CISO and IR lead within 40 minutes, then a one-page brief to the executive team within four hours
Feedback	Post-incident review produced two new requirements and three new detection rules

The flash report, as written:

We assess with **high confidence** that an active intrusion is in progress, consistent with the pre-encryption stages of a known ransomware playbook. Three behaviors observed in the last 72 hours match the documented sequence, including authentication to backup infrastructure, which in this playbook immediately precedes encryption. We assess encryption is **likely** within 24 to 72 hours. Recommend immediate isolation of the identified workstation and service account, out-of-band verification of backup integrity, and activation of the incident response plan.

Read that again and count what it contains: a judgment, a confidence level, the specific basis, a forecast with a time window, an estimative probability term, and three specific assignable actions. It is four sentences.

How the outcome improved.

- Encryption was prevented. Isolation occurred approximately 30 hours before the assessed window.
- Backups were verified as intact and were confirmed offline, which removed the attacker's primary leverage.
- The fraudulent vendor payment change was reversed before the payment cycle ran.

- Because the attacker's method was understood, containment addressed the access path rather than only the visible symptom.
- Detection rules built from the incident later caught a second, unrelated intrusion attempt.

The counterfactual. Without the strategic and operational intelligence layers, that alert is closed as a false positive, consistent with the previous 60. Encryption follows within days. Clinical systems go offline. Recovery in comparable published incidents commonly runs weeks and costs run to millions, with patient care diverted in the interim.

The teaching point. Notice that the tactical indicators alone would not have saved this organization. The indicator feed was already available and largely unused. What made the difference was the **operational** understanding of the playbook, which told the analyst what the alert *meant in sequence*, and the **strategic** assessment, which had made this threat a standing priority worth writing a requirement for. This is the single clearest illustration in this chapter of why the three levels are not optional alternatives.

Discussion questions.

1. Which lifecycle phase was most responsible for the successful outcome? Defend your answer.
 2. The flash report claimed high confidence from three correlated behaviors. Was that justified? What would have made it moderate instead?
 3. The organization had one part-time intelligence analyst. Write the business case for making that role full time, using this incident.
-

Case Study 5: Emergency Management

The Ridge County Wildfire Evacuation (composite, informed by documented practice)

Composite case. The analytic patterns and the lessons about pre-decision are drawn from documented emergency management practice.

The situation. Ridge County contains two subdivisions, roughly 1,800 households combined, on the downslope side of a ridge. Both are served by a single two-lane access road. County emergency management has a persistent problem: evacuation decisions have historically been made at ignition, and the road cannot clear 1,800 vehicles in the time available once a fire crosses the ridge.

The requirement. The county emergency manager needed to answer two questions before fire season, not during it:

- Strategic: what is our actual evacuation capacity for these subdivisions, and under what conditions is it insufficient?
- Operational: what set of observable conditions should trigger a pre-emptive advisory, and who has authority to issue it?

Disciplines and information sources used.

Source	Contribution
MASINT-adjacent sensing	Weather station data: wind speed and direction, temperature, relative humidity, plus fuel moisture sampling
GEOINT	Terrain and slope modeling, fuel load mapping, structure locations, road network capacity analysis
Historical analysis	Reconstruction of the two most destructive fires in county history, including hourly spread rates and the conditions present at ignition
OSINT	Regional and national fire weather forecasting products
HUMINT	Debriefs of the fire officers who worked the two historical fires, capturing operational detail never written into the after-action reports
Internal data	Traffic modeling of the access road, plus a household survey establishing vehicles per household and estimated departure lag

The analysis. Two findings, both quantitative and both uncomfortable.

1. **A condition threshold.** Both historical fires crossed the ridge within six hours of ignition, and both ignited on days with sustained winds above roughly 15 mph from the northeast, relative humidity below 12%, and fuel moisture below a specific measured level. When any two of those three conditions were present without the third, historical fires had not crossed within six hours.
2. **An evacuation capacity gap.** Traffic modeling showed the single access road clearing approximately 1,100 vehicles in four hours under realistic departure-lag assumptions. Expected vehicle demand was approximately 2,600. The gap was not marginal. Ordering evacuation at ignition would strand roughly 1,500 vehicles on a road with fire approaching.

The lifecycle in action.

- **Planning:** two requirements, one strategic and one operational, both tied to a specific decision authority.
- **Collection:** a full pre-season effort across sensors, terrain modeling, historical files, practitioner interviews, and a household survey.
- **Processing:** historical fire records converted into a consistent hourly timeline with standardized weather readings, which had never been done because each fire was documented separately in a different format.
- **Analysis:** threshold identification and capacity modeling, with the key assumptions written

down, including departure lag, which was the assumption the whole model was most sensitive to.

- **Dissemination:** a decision matrix, one page, laminated, in every relevant vehicle and dispatch center. Not a report. A matrix that names the conditions and the action.
- **Feedback:** the matrix was exercised in a tabletop drill before fire season, which revealed that the authority to issue a voluntary advisory was ambiguous. That ambiguity was resolved in writing before it mattered.

The decision matrix, simplified:

Conditions present	Action	Authority
All three threshold conditions forecast within 24 hours	Pre-stage engines, notify subdivisions, open the alternate egress gate, issue voluntary advisory	Emergency manager, no further approval needed
All three present and any ignition reported in the sector	Mandatory evacuation immediately, do not wait for spread confirmation	Emergency manager or fire chief, either alone
Two of three conditions	Heightened watch, staff dispatch, verify contact lists	Duty officer
Ignition with fewer than two conditions	Standard response	Duty officer

How the outcome improved. In August, all three conditions were forecast. The county issued a voluntary advisory and opened the alternate egress route 19 hours before an ignition occurred in the sector. Roughly 60% of households had already left when the mandatory order was issued. The remaining demand was within the road's capacity. Evacuation completed with no fatalities and no vehicles stranded on the access road.

The critical point: **nothing about the fire was different.** The fire behaved as the historical analysis predicted. What changed was that the decision had been pre-made, the threshold had been defined in advance, and the authority to act on it was unambiguous. That is the characteristic contribution of intelligence to emergency management. It moves decisions earlier, when options are still available.

The counterfactual. Ordering evacuation at ignition puts approximately 2,600 vehicles onto a road that can move 1,100 in the available window. This is the pattern behind a number of the deadliest wildfire evacuation failures on record: not an absence of information, but a decision made at the moment of the event rather than before it.

Discussion questions.

1. Which finding was more valuable, the condition threshold or the capacity gap? Could either stand alone?
2. The model was most sensitive to the departure-lag assumption. How would you test that

assumption before relying on it?

3. The tabletop exercise exposed an authority ambiguity. Which lifecycle phase does an exercise belong to, and why does that matter?

Cross-Case Comparison

Case	Primary disciplines	Dominant level	Decision improved	Core lesson
Cuban Missile Crisis	IMINT, HUMINT	Strategic with tactical urgency	Graduated response instead of a binary choice	A well-formed requirement plus the right discipline creates options
Meridian Foods	OSINT, GEOINT, TECHINT, ethical HUMINT, internal BI	Strategic informing operational	Resources redirected before the competitive event	Leading indicators beat lagging reports
Boston focused deterrence	Records analysis, practitioner HUMINT, network analysis	Operational	Precision replaced volume	Analysis reveals concentration, and concentration is where leverage is
Cascade Regional Health	Strategic, operational, and tactical CTI plus owned telemetry	All three, working together	Prevention instead of recovery	Tactical indicators are useless without operational context
Ridge County wildfire	GEOINT, sensing, historical analysis, practitioner HUMINT	Strategic producing an operational trigger	The decision moved 19 hours earlier	Pre-deciding is the highest-value output intelligence can produce

Look down the "core lesson" column. Five different domains, and the same underlying claim in every one: the value of intelligence is not in knowing more. It is in deciding better and earlier.

Part 4: Chapter Summary

Information and intelligence are different things. Information is a collection of facts that exists whether or not anyone needs it. Intelligence is information collected against a stated requirement, evaluated for reliability, analyzed in context, given a judgment and a confidence level, and delivered to a specific decision maker in time to matter. Remove any of those elements and you are back to information. Volume does not close the gap; only analysis does.

The progression from data to intelligence is deliberate work. Data becomes information when

context is added, information becomes knowledge when patterns and comparison are added, and knowledge becomes intelligence when a human adds judgment, forecast, and accountability. Each step requires a person willing to own a conclusion.

Evaluation has two independent parts. Source reliability asks whether we trust the provider. Information credibility asks whether this specific report is plausible. Reliable sources can be wrong and unreliable sources can be right, so the two must be judged separately and reported separately.

The intelligence lifecycle has six phases. Planning and direction defines the requirement. Collection gathers raw information. Processing and exploitation makes it usable. Analysis and production turns it into a judgment. Dissemination delivers it in a usable form to the right person. Feedback and evaluation closes the loop and generates new requirements. In practice the phases overlap and run in parallel, but skipping one always costs something, and the two most commonly skipped are planning (which produces unanswerable questions) and feedback (which produces an organization that cannot learn).

Intelligence disciplines are defined by source, not topic. OSINT collects publicly available material and carries most of the load in most assessments. HUMINT collects from people and is uniquely capable of reaching intent. SIGINT collects intercepted signals and, for private actors, is largely off limits. IMINT and GEOINT collect images and geospatial data and provide physical proof. MASINT measures physical signatures. TECHINT examines the adversary's or competitor's hardware. FININT follows money. CYBINT and CTI cover digital threat activity. Single-discipline assessments are fragile; all-source fusion is the professional standard, and confidence rises most when the corroborating disciplines would be hard to manipulate together.

Strategic, operational, and tactical are levels, not types. Strategic intelligence serves long-horizon direction-setting for senior leadership. Tactical intelligence serves immediate action by operators and is highly perishable. Operational intelligence sits between them, describing structures, networks, and campaigns so that coordinated action can be planned. Operational is the layer most often missing, and its absence is why some organizations stay permanently busy without getting ahead of anything.

Competitive intelligence is a domain, and it is bounded by ethics and law. It covers competitors, customers, suppliers, new entrants, and substitutes, and it contains strategic, operational, and tactical levels within itself. What separates it from industrial espionage is entirely a matter of collection method. If you would have to misrepresent your identity, or defeat measures someone took to keep information secret, you are outside the profession.

Criminal intelligence orients toward pattern and structure, not just cases. Intelligence-led policing directs resources toward the concentrations of place, offender, and victim that analysis reveals. Its core products are pattern analysis, hot spot mapping, link and network analysis, target profiles, and threat and problem assessments. It operates under constraints that business intelligence does not face, including criminal predicate requirements, retention limits, audit obligations, and the separation of intelligence from evidence.

Business intelligence is inward-looking analytics infrastructure. It integrates and presents the organization's own data. It answers what happened and why, and it is essential, but it is not a substitute for outward-looking competitive intelligence, and BI stacks routinely omit the analysis layer that would

turn dashboards into decisions.

Good products share a shape. Bottom line up front. Facts separated from assessments. Confidence stated with the reason for it. Assumptions written down. Gaps named. Recommendations specific and assignable. Next collection identified.

The measure of intelligence is not accuracy alone. It is whether a decision changed for the better. A correct assessment that nobody read failed. That standard is what distinguishes intelligence as a profession from research as an activity.

Part 5: Key Terms

Terms are grouped by theme rather than alphabetized, because related terms are easier to learn together. An alphabetical index follows the definitions.

5.1 Foundational Terms

Data. Raw, unorganized values, symbols, or measurements with no attached meaning. "4,412" is data.

Information. Data that has been organized, labeled, and placed in enough context that a human can understand what it describes. Information exists independently of anyone needing it.

Knowledge. Information that has been compared against history, baselines, and other information so that patterns become visible.

Intelligence. Information that has been collected against a stated requirement, evaluated for reliability, analyzed in context, assigned a judgment and a confidence level, and delivered to a specific decision maker in time to affect a decision.

Actionable intelligence. Intelligence specific enough that the recipient knows what to do differently as a result. If the reader finishes it and has no changed course of action available, it was not actionable.

Decision advantage. The condition of being able to decide faster or better than an adversary or competitor. The purpose of intelligence, stated in one phrase.

Intelligence product. Any written or briefed output of the analysis phase. A report, brief, bulletin, assessment, or watchlist.

Consumer. The person the product is written for. Also called the customer or the decision maker.

All-source analysis. Analysis that fuses multiple collection disciplines into a single assessment, so that the weaknesses of each discipline are covered by the others.

Estimative language. Standardized probability vocabulary (likely, unlikely, almost certainly) used so that readers interpret an analyst's uncertainty consistently.

Confidence level. A stated judgment (high, moderate, low) about how solid the basis for an assessment is. Distinct from probability: an analyst can have high confidence that an event is unlikely.

BLUF. Bottom Line Up Front. The convention of putting the judgment in the first sentence of a product.

5.2 Lifecycle Terms

Intelligence lifecycle (intelligence cycle). The repeating process of planning and direction, collection, processing and exploitation, analysis and production, dissemination, and feedback.

Planning and direction. The phase in which decision requirements are converted into specific, answerable intelligence requirements and a collection plan.

Intelligence requirement. A specific question that must be answered to support a decision.

Priority Intelligence Requirement (PIR). A requirement the decision maker has personally endorsed as a priority, which drives resource allocation.

Essential Elements of Information (EEI). The specific sub-questions that must be answered in order to satisfy a PIR.

Indicator. An observable thing whose presence or absence helps answer a requirement. Indicators are defined in advance, not recognized afterward.

Intelligence gap. A specific thing the analyst does not know, stated explicitly, along with how resolving it would change the assessment.

Collection plan. A document assigning each information gap to a specific source or discipline, with timing.

Collection. The deliberate gathering of raw information against the plan.

Processing and exploitation. Converting raw collection into usable form: translation, transcription, decryption where lawful, normalization, deduplication, entity resolution, and indexing.

Normalization. Converting data into consistent formats so it can be compared. Standardizing time zones, currencies, name formats, and date conventions.

Entity resolution. Determining that multiple references ("R. Smith," "Bob Smith," "Robert A. Smith") describe one real-world entity, or that seemingly identical references describe different entities.

Analysis and production. The phase in which evaluated information is integrated, judged, and written into a product.

Dissemination. Delivering the finished product to the right consumer in the right format, through the right channel, at the right time, subject to handling controls.

Feedback and evaluation. The phase in which the consumer reports whether the product was useful, accuracy is assessed after the fact, and new requirements are generated.

Need to know. The principle that access to sensitive material is granted based on a demonstrated requirement, not on rank or curiosity.

5.3 Source and Evaluation Terms

Source reliability. An assessment of how much the provider of information can be trusted, based on track record. Commonly rated A through F.

Information credibility. An assessment of how plausible a specific report is on its own merits, independent of who provided it. Commonly rated 1 through 6.

Corroboration. Independent confirmation of a report by a separate source that did not derive from the first.

Circular reporting. The false appearance of corroboration created when several apparently independent sources all trace back to one original, unverified report. A dominant risk in open source work.

Single-source dependency. Reliance on one source for a critical judgment, which makes the entire assessment vulnerable to that source being wrong or manipulated.

Deception. Deliberate provision of false information intended to mislead an analyst.

Denial. Deliberate concealment of true information to prevent collection. Denial and deception are usually discussed together as D&D.

Elicitation. Obtaining information through ordinary-seeming conversation, without the subject realizing they are being questioned. A HUMINT technique and, correspondingly, a counterintelligence concern.

5.4 Discipline Terms

Intelligence discipline. A category of collection defined by the type of source and method, not by subject matter.

OSINT. Open Source Intelligence. Collection from publicly and lawfully available material.

SOCMINT. Social Media Intelligence. A sub-practice of OSINT focused on social platforms.

HUMINT. Human Intelligence. Collection from people through deliberate interaction.

SIGINT. Signals Intelligence. Collection from intercepted signals, comprising COMINT (communications), ELINT (electronic emissions), and FISINT (foreign instrumentation).

IMINT. Imagery Intelligence. Collection from satellite, aerial, drone, and ground imagery.

GEOINT. Geospatial Intelligence. The broader discipline combining imagery with mapping, terrain, and other geographically referenced data.

MASINT. Measurement and Signature Intelligence. Collection from physical signatures: radiation, acoustic, seismic, chemical, thermal, magnetic.

TECHINT. Technical Intelligence. Intelligence derived from examining an adversary's or competitor's equipment or products, including reverse engineering and teardown analysis.

FININT. Financial Intelligence. Intelligence derived from financial transactions, records, and flows.

CYBINT / CTI. Cyber Intelligence / Cyber Threat Intelligence. Intelligence about threat actors, tools, infrastructure, and campaigns in the digital domain.

Indicator of Compromise (IOC). A specific technical artifact (file hash, IP address, domain, registry key) associated with malicious activity. Tactical-level CTI.

Change detection. Comparison of imagery or data of the same location or system across time to identify what has changed. A core GEOINT method.

5.5 Level Terms

Strategic intelligence. Intelligence supporting long-horizon decisions about direction, posture, and resource commitment, consumed by senior leadership.

Operational intelligence. Intelligence supporting the planning and management of campaigns, cases, and programs over weeks to a year, describing structures and processes.

Tactical intelligence. Intelligence supporting immediate action, highly specific, highly perishable, consumed by operators.

Perishability. The rate at which intelligence loses value as time passes. High in tactical products, low in strategic ones.

Time horizon. The span of future time an intelligence product addresses.

5.6 Domain Terms

Competitive intelligence (CI). The lawful and ethical collection and analysis of information about competitors, customers, suppliers, entrants, and substitutes, to support business decisions.

Business intelligence (BI). The technologies and practices used to integrate, analyze, and present an organization's own internal data for management decision support.

Criminal intelligence. The product of analyzing information about criminal activity, offenders, and networks to prevent, disrupt, and prosecute crime and allocate resources.

Intelligence-led policing (ILP). A policing model in which enforcement resources are directed by analysis of concentrations in place, offender, and victim rather than by reactive volume alone.

Industrial espionage. Unlawful acquisition of another organization's protected information, including through misrepresentation, trespass, unauthorized system access, or inducing breach of confidentiality obligations.

Counterintelligence. Activity intended to identify, assess, and neutralize another party's intelligence collection against you. In business, includes footprint auditing and elicitation training.

Trade secret. Information that has commercial value because it is not generally known, and which its owner has taken reasonable measures to keep secret. The "reasonable measures" element is why defeating protective measures is legally significant.

Link analysis. A method of charting relationships among people, organizations, locations,

communications, and assets to reveal structure.

Social network analysis. Quantitative analysis of relationship structures, identifying central actors, brokers between groups, and structural choke points.

Hot spot analysis. Geographic analysis identifying where incidents concentrate.

Criminal predicate. A reasonable suspicion of criminal involvement, required in many jurisdictions before an individual may be the subject of retained intelligence records.

Focused deterrence. A crime reduction strategy in which analysis identifies specific groups driving violence, who then receive a direct and credible warning alongside offers of services and support.

Fusion center. A facility where multiple agencies share and jointly analyze information.

Data warehouse. A centralized repository of integrated data structured for analysis rather than for transaction processing.

Semantic layer. The definitional layer in a BI stack that establishes what each metric means across the organization.

Descriptive, diagnostic, predictive, prescriptive analytics. A maturity progression answering, respectively, what happened, why it happened, what will happen, and what should be done.

5.7 Analytic Method and Bias Terms

Structured analytic technique. A named, repeatable method used to reduce the influence of bias and to test reasoning.

Analysis of Competing Hypotheses (ACH). A technique that lists all plausible explanations and evaluates evidence against each, emphasizing disconfirmation rather than confirmation.

Key Assumptions Check. A technique that lists every assumption an assessment rests on and tests what happens if each is wrong.

Devil's advocacy. Assigning a person or team to argue the opposing case in good faith.

Red teaming. Modeling an adversary's decision making from the adversary's own perspective and incentives.

Indicators and warnings (I&W). Defining in advance the observables that would signal a change, then monitoring for them.

Confirmation bias. The tendency to notice and credit evidence supporting an existing belief while discounting contrary evidence.

Mirror imaging. Assuming an adversary reasons, values, and prioritizes the way the analyst does.

Anchoring. Excessive reliance on the first piece of information received.

Layering. Treating an assumption or judgment from an earlier product as established fact in a later one, so that uncertainty silently disappears over successive products.

Politicization. Adjusting an analytic judgment to match what the consumer wants to hear. The most damaging analytic failure, because it destroys the value of all future products.

Groupthink. Convergence of a team on a single view through social pressure rather than evidence.

Vanity metric. A measure that reliably improves and never informs a decision.

5.8 Alphabetical Index of Key Terms

ACH · Actionable intelligence · All-source analysis · Anchoring · Analysis and production · BI · BLUF · Business intelligence · Change detection · CI · Circular reporting · COMINT · Collection · Collection plan · Competitive intelligence · Confidence level · Confirmation bias · Consumer · Corroboration · Counterintelligence · Criminal intelligence · Criminal predicate · CTI · CYBINT · Data · Data warehouse · Deception · Decision advantage · Denial · Descriptive analytics · Devil's advocacy · Diagnostic analytics · Dissemination · EEI · Elicitation · ELINT · Entity resolution · Estimative language · Feedback and evaluation · FININT · FISINT · Focused deterrence · Fusion center · GEOINT · Groupthink · Hot spot analysis · HUMINT · I&W · ILP · IMINT · Indicator · Indicator of compromise · Industrial espionage · Information · Information credibility · Intelligence · Intelligence discipline · Intelligence gap · Intelligence lifecycle · Intelligence product · Intelligence requirement · Intelligence-led policing · Key assumptions check · Knowledge · Layering · Link analysis · MASINT · Mirror imaging · Need to know · Normalization · Operational intelligence · OSINT · Perishability · PIR · Planning and direction · Politicization · Predictive analytics · Prescriptive analytics · Processing and exploitation · Red teaming · Semantic layer · SIGINT · Single-source dependency · Social network analysis · SOCMINT · Source reliability · Strategic intelligence · Structured analytic technique · Tactical intelligence · TECHINT · Time horizon · Trade secret · Vanity metric

Part 6: Review Questions

6.1 Short Answer (20 questions)

Answer in three to five sentences unless otherwise noted. Full-credit answers are in the Answer Key at the end of the workbook.

1. Define information and intelligence, and name the five elements that must be present for something to qualify as intelligence.
2. Explain why collecting more data does not automatically produce better intelligence. Give one concrete example.
3. Describe the four rungs of the data-to-intelligence ladder and identify what each rung adds that the one below it lacks.
4. Distinguish source reliability from information credibility, and explain why they must be assessed separately.

5. What does a report rated B2 tell you? What does F3 tell you?
6. List the six phases of the intelligence lifecycle in order and state the purpose of each in one sentence.
7. Explain the relationship among a decision, a PIR, an EEI, and an indicator. Use an example.
8. Give the four tests of a well-formed intelligence requirement.
9. Why is processing and exploitation described as invisible when it works and catastrophic when it fails? Give two specific failure examples.
10. Explain the difference between a confidence level and a probability estimate. Give an example sentence that uses both correctly.
11. Name three structured analytic techniques and state what problem each is designed to counter.
12. Why is feedback the most commonly skipped phase, and what is the specific cost of skipping it?
13. Explain the BLUF convention and why academic writing structure is wrong for intelligence products.
14. Define intelligence discipline. Explain why "counterterrorism" is not a discipline.
15. Compare the strengths and weaknesses of OSINT and HUMINT. Under what circumstances is HUMINT worth its higher cost and risk?
16. Distinguish strategic, operational, and tactical intelligence by time horizon, consumer, and product type.
17. Why is operational intelligence the layer most often missing in real organizations, and what symptom does its absence produce?
18. State the identity rule and the publication rule for competitive intelligence, and explain what each is designed to prevent.
19. Explain the difference between business intelligence and competitive intelligence, and why an organization with excellent BI can still be blindsided.
20. Using the Cascade Regional Health case, explain why tactical indicators alone would not have prevented the ransomware event.

6.2 Multiple Choice (15 questions)

Select the single best answer.

1. Which element is present in intelligence but absent from information? a) Accuracy b) An explicit judgment with a stated confidence level c) A large volume of supporting data d) A published source citation
2. A police department installs 200 new cameras but adds no analysts. The most likely result is: a) A proportional increase in usable intelligence b) Better strategic assessments c) An increase in collected

data with little change in intelligence produced d) Automatic identification of crime patterns

3. A source has been accurate 30 times consecutively but this particular report contradicts three other reliable sources. The most appropriate evaluation is: a) High reliability, high credibility b) High reliability, low credibility c) Low reliability, high credibility d) The report should be discarded without evaluation

4. The intelligence lifecycle phase that most often causes total cycle failure when performed poorly is: a) Collection b) Processing c) Planning and direction d) Dissemination

5. Which of the following is an Essential Element of Information rather than a Priority Intelligence Requirement? a) Will the competitor enter our market within 18 months? b) Should we commit \$12M to a new distribution center? c) Has the competitor posted regional driver job openings? d) What is our overall competitive position?

6. Normalizing timestamps across three sources belongs to which phase? a) Collection b) Processing and exploitation c) Analysis and production d) Dissemination

7. "We assess with high confidence that an attack is unlikely" is: a) Self-contradictory and should be rewritten b) A coherent statement, because confidence and probability are different measures c) A statement of low confidence d) An example of politicization

8. Which discipline is best suited to determining an adversary's intent rather than their capability? a) IMINT b) MASINT c) HUMINT d) TECHINT

9. A company purchases a competitor's product at retail and disassembles it to analyze cost structure. This is: a) Industrial espionage b) TECHINT, and normal lawful practice c) SIGINT d) A violation of trade secret law

10. Three online sources report the same claim, and all three cite a single unverified original post. This is an example of: a) Corroboration b) Circular reporting c) All-source analysis d) Change detection

11. A blocklist of 14 domains with a detection rule is best described as: a) Strategic intelligence b) Operational intelligence c) Tactical intelligence d) Business intelligence

12. A network chart showing a trafficking organization's roles, relationships, and enforcement choke points is primarily: a) Strategic intelligence b) Operational intelligence c) Tactical intelligence d) Competitive intelligence

13. Which activity crosses the line from competitive intelligence into industrial espionage? a) Analyzing a competitor's job postings by metro area b) Attending a competitor's public conference presentation c) Posing as a recruiter to extract confidential roadmap details from an employee d) Purchasing commercial satellite imagery of a competitor's facility

14. An analyst assumes a foreign government will respond to sanctions the way a Western government would. This is: a) Anchoring b) Mirror imaging c) Layering d) Circular reporting

15. In the Ridge County wildfire case, the highest-value contribution of the intelligence work was: a) Predicting the exact ignition point b) Increasing the number of weather sensors c) Moving the evacuation decision earlier by defining a trigger in advance d) Producing a longer and more detailed

report

6.3 True or False (10 questions)

Mark each statement true or false. For any statement you mark false, write one sentence explaining what makes it false.

1. Intelligence and information mean essentially the same thing in professional practice.
 2. A perfectly accurate analysis delivered after the decision has been made still qualifies as intelligence.
 3. Source reliability and information credibility are two names for the same assessment.
 4. The intelligence lifecycle in real practice is a clean sequential circle with no overlapping phases.
 5. Open source intelligence is generally the lowest-cost and lowest-legal-risk discipline available to private sector analysts.
 6. Tactical intelligence is usually the most perishable of the three levels.
 7. Competitive intelligence and industrial espionage are distinguished primarily by the analytic methods used.
 8. Business intelligence looks outward at competitors, while competitive intelligence looks inward at internal data.
 9. A confidence level tells the reader how solid the basis for a judgment is, not how likely the event is.
 10. In most democratic legal systems, retaining intelligence records on an individual generally requires some form of criminal predicate.
-

Part 7: Critical Thinking Exercises

For each scenario: identify the type or types of intelligence in use, identify the level (strategic, operational, or tactical), identify the disciplines involved, and explain your reasoning in three to six sentences. Several scenarios are deliberately ambiguous or contain a mistake for you to catch.

Exercise 7.1. A hospital system's board commissions a study on which categories of threat actor are most likely to target regional healthcare providers over the next three years, and what capability the system should build in response.

Exercise 7.2. At 03:12, a SOC analyst receives a pushed alert listing a file hash, the affected endpoint, and the instruction to isolate the host immediately.

Exercise 7.3. A detective sergeant reviews a chart showing that one wholesale supplier connects to nine street-level sellers across three neighborhoods, and recommends sequencing the case to target the supplier and an associated cash business.

Exercise 7.4. A regional sales director receives a monthly report showing revenue by product line and territory for the previous quarter, with variance against forecast.

Exercise 7.5. A retail chain analyzes commercial satellite imagery of its competitor's parking lots across 40 locations weekly, counting vehicles to estimate the competitor's traffic trend.

Exercise 7.6. A county emergency manager builds a laminated decision matrix specifying which weather conditions trigger a voluntary evacuation advisory, and who has authority to issue it.

Exercise 7.7. An analyst writes: "The subject's armament status is presently indeterminate pending further collection." The product is a pre-entry briefing for a tactical team executing a warrant in 20 minutes. Identify the problem and rewrite the sentence.

Exercise 7.8. A consultant offers to sell a company a list of 90,000 malicious IP addresses updated hourly, described as "complete cyber threat intelligence." Assess this claim.

Exercise 7.9. An analyst covering a foreign military notices that reports supporting an imminent-mobilization hypothesis feel compelling, while reports suggesting a routine exercise feel weak and are set aside without documentation. Name the failure and prescribe a specific technique to counter it.

Exercise 7.10. A company's competitive intelligence team produces an assessment based partly on a document a new hire brought from their previous employer, marked confidential. The analysis is excellent. Assess the situation.

Exercise 7.11. An analyst's assessment three products ago stated, with low confidence, that a competitor was probably developing a subscription product. The current product states as background: "The competitor's subscription product, currently in development..." Name the failure.

Exercise 7.12. A CISO asks for "everything we have on ransomware." Rewrite this as a well-formed requirement, and explain what makes your version better.

Exercise 7.13. A city deploys a system that forecasts which blocks will see the most property crime, trained on ten years of arrest data. Patrol is allocated accordingly. Identify the technical and civil rights concerns and describe two safeguards.

Exercise 7.14. A company has 340 dashboards, four analytics engineers, and no one who can name a decision that changed last quarter as a result of any dashboard. Diagnose the problem using the vocabulary of this chapter, and state what you would add.

Exercise 7.15. An emergency manager receives an assessment stating that a levee has a "meaningful probability of failure under sustained rainfall exceeding historical norms." Explain why this product is not usable and specify exactly what is missing.

Part 8: Hands-On Lab Exercises

General Lab Guidance

Read before starting any lab.

All five labs use only public information and your own judgment. No lab requires you to access a system you do not own, contact a person under a false identity, or collect information about a private individual. If any lab appears to require one of those things, you have misread the lab. Stop and re-read.

Rubric scale used throughout. Each rubric criterion is scored 0 to 4:

Score	Meaning
4	Professional quality. Could be submitted to a real decision maker with no changes.
3	Competent. Meets requirements with minor weaknesses.
2	Developing. Meets some requirements; significant gaps in reasoning or format.
1	Attempted. Shows awareness of the concept but does not apply it correctly.
0	Missing or fundamentally incorrect.

Submission format for all labs. Typed, with your name, the lab number, and the date. Worksheets may be handwritten and scanned. Every judgment you write must carry a confidence level. Every product must name at least one intelligence gap. Those two requirements are graded in every lab.

Lab 1: Converting Raw Information Into Actionable Intelligence

Time required: 90 to 120 minutes **Format:** Individual **Difficulty:** Foundational

Objectives

By the end of this lab you will be able to:

1. Classify individual items as data, information, or intelligence.
2. Assign source reliability and information credibility ratings and defend them.
3. Identify which items are relevant to a stated requirement and which are noise.
4. Produce a written intelligence summary with a judgment, a confidence level, stated assumptions, named gaps, and specific recommendations.

Prerequisites

- Sections 1.1, 1.2, and 1.10 of the reading material.
- Familiarity with the Intelligence Summary template in Section 1.10.
- No software required. No internet access required.

Scenario

You are the sole intelligence analyst for **Harborline Logistics**, a mid-sized regional freight company with 240 trucks and four terminals. Your operations director has given you a requirement:

PIR: Is our largest customer, Brightwell Manufacturing (18% of our revenue), preparing to move its freight business to another carrier? If so, on what timeline?

Decision it supports: Whether to commit an account team and a rate concession to retain Brightwell before their contract renewal in 90 days.

Raw Reporting Provided

You have the following fifteen items. Some matter. Some do not.

#	Item	Provenance
1	Brightwell's shipment volume with us fell 11% in the last 60 days.	Our own operational data
2	Brightwell posted a job opening for a "Carrier Relations Manager, multi-carrier environment."	Public job board
3	A driver mentioned seeing two trucks from a competitor at Brightwell's dock last Tuesday.	Verbal, secondhand, driver has been reliable on operational matters for six years
4	Brightwell's quarterly earnings release stated a company-wide goal of reducing logistics cost per unit by 8%.	Public investor release
5	Brightwell's plant manager cancelled a scheduled quarterly business review with us, citing scheduling.	Our account manager
6	An anonymous post on an industry forum claims "a big Midwest manufacturer is about to dump its incumbent carrier."	Anonymous, unverifiable
7	Brightwell's total production volume is up 4% year over year, per their earnings release.	Public investor release
8	Our on-time delivery rate for Brightwell fell from 96% to 88% over the last two quarters.	Our own operational data
9	Brightwell's new VP of Supply	Public professional profile

#	Item	Provenance
	Chain came from a company that used two of our competitors.	
10	A competitor's regional sales rep was overheard at a trade show saying his team was "working a big account in this region."	Our sales rep, firsthand, but the competitor rep is a known exaggerator
11	Brightwell's headquarters cafeteria was renovated last month.	Local news
12	Brightwell issued an RFP for warehousing services (not freight) 30 days ago.	Public procurement notice
13	Fuel surcharges industry-wide rose 6% this quarter.	Trade publication
14	Our billing department reports three disputed invoices from Brightwell in 60 days, versus one in the prior year.	Our own records
15	Brightwell's CEO sold a routine block of shares under a pre-scheduled plan.	Public filing

Step-by-Step Instructions

Step 1 (10 min). Read the PIR aloud. Write, in your own words, exactly what decision this intelligence must support and what deadline applies. Do not proceed until you can state it in one sentence.

Step 2 (15 min). Complete Worksheet 1A. For each of the fifteen items, classify it as **Data**, **Information**, or **Knowledge**. None of the fifteen items is intelligence yet. Be prepared to explain why.

Step 3 (15 min). Complete Worksheet 1B. Assign each item a reliability letter (A through F) and a credibility number (1 through 6). Write one short reason for each rating.

Step 4 (10 min). Complete Worksheet 1C. Mark each item as **Relevant**, **Possibly relevant**, or **Noise** with respect to the PIR. You should find at least three items that are noise. Identify them and say why they are tempting.

Step 5 (15 min). Group the relevant items into **indicator categories**. Suggested categories: behavioral change by the customer, capability building by the customer, our own performance problems, competitor activity, and market-wide factors that affect everyone. Note which category our own performance falls into, and think about what that means.

Step 6 (10 min). Write down at least two competing hypotheses that explain the relevant items. One should be "Brightwell is preparing to switch carriers." Write at least one plausible alternative. Then, for

each item, mark whether it supports, contradicts, or is neutral toward each hypothesis. Look for items that discriminate between the hypotheses. Most items will not discriminate, and recognizing that is the point of this step.

Step 7 (25 min). Write the intelligence summary using the template from Section 1.10. Constraints:

- Bottom line, no more than three sentences, with a confidence level.
- Exactly five key facts, each with its reliability/credibility rating.
- Assessment paragraph that names the alternative hypothesis and says why you judged it less likely.
- At least two key assumptions, each with a statement of what breaks if it is wrong.
- At least two intelligence gaps.
- Three recommended actions, each specific, assignable, and time-bound.
- Two collection requests.
- Total length: one page maximum. This limit is graded.

Step 8 (10 min). Self-review against this checklist before submitting:

- Does my first sentence contain a judgment and a confidence level?
- Could a reader who reads only my first paragraph act correctly?
- Have I separated facts from assessment, visibly?
- Is every recommendation something a specific person could start tomorrow?
- Did I include any item I know to be noise?

Worksheet 1A: Ladder Classification

Item #	Data / Information / Knowledge	Why
1		
2		
3		
...		
15		

Worksheet 1B: Source Evaluation

Item #	Reliability (A-F)	Credibility (1-6)	Justification
1			

Item #	Reliability (A-F)	Credibility (1-6)	Justification
...			
15			

Worksheet 1C: Relevance Triage

Item #	Relevant / Possibly / Noise	Reasoning
1		
...		
15		

Worksheet 1D: Competing Hypotheses Matrix

Item #	H1: Preparing to switch	H2: (your alternative)	Discriminating?
1	Supports / Contradicts / Neutral		Yes / No
...			

Expected Outcomes

A competent submission will:

- Correctly identify items 11, 13, and 15 as noise. Item 11 is irrelevant. Item 13 is market-wide and affects all carriers equally, so it does not discriminate. Item 15 is a routine pre-scheduled transaction with no bearing on freight decisions.
- Rate item 6 as low reliability and low credibility, and either exclude it or include it with explicit caveats. It is anonymous, unverifiable, and could describe any of hundreds of companies.
- Recognize that item 8 (our own declining on-time performance) is the most important item in the set, because it is the only one that describes a cause we control. Strong submissions will note that this reframes the recommendation from "concede rate" to "fix service."
- Rate items 1, 8, and 14 as A-grade reliability, since they come from our own records.
- Assign moderate confidence overall. High confidence is not defensible from this reporting; low confidence understates the convergence of multiple independent indicators.
- Name at least one alternative hypothesis, most commonly that Brightwell is adding a second carrier for capacity or cost-benchmarking purposes rather than replacing us, which is consistent with items 2, 4, and 7 and would call for a different response.

Discussion Questions

1. Item 8 is our own failure. How does the analytic product change when the cause is internal rather than external, and what pressure does that create on the analyst?
2. Items 3 and 10 are both hearsay from people we know. Why do they deserve different ratings?
3. If your assessment led the company to concede a rate reduction and Brightwell had never intended to leave, what was the cost of your error? Now consider the reverse error. Which is worse, and does that change how you should write the confidence statement?
4. You were given fifteen items. In real practice, you would have hundreds. What process would you use to triage at that scale, and what would you lose?

Grading Rubric: Lab 1 (44 points)

Criterion	Weight	4 points looks like
Ladder classification accuracy (1A)	x2	Nearly all items correctly classified with sound reasoning
Source evaluation quality (1B)	x2	Ratings defensible, and internally consistent; own-records rated appropriately high
Noise identification (1C)	x2	All three noise items identified with correct reasoning
Competing hypotheses (1D)	x2	A genuine, plausible alternative developed and tested, not a straw man
BLUF quality	x1	Judgment and confidence in the first sentence, actionable on its own
Fact / assessment separation	x1	Reader can tell at a glance what was observed versus concluded
Confidence calibration and justification	x2	Level assigned is defensible and the reason for it is stated
Assumptions and gaps	x1	At least two of each, with consequences stated
Recommendations	x1	Specific, assignable, time-bound, and connected to the assessment
Length discipline (one page)	x1	Met without sacrificing required content

Total possible: 44. Passing: 30.

Lab 2: Mapping the Complete Intelligence Lifecycle

Time required: 2 to 3 hours **Format:** Individual or pairs **Difficulty:** Intermediate

Objectives

1. Decompose a real-world decision into a PIR, EEIs, and indicators.
2. Build a collection plan that assigns each gap to a specific discipline and source.
3. Identify the processing steps required to make collected material comparable.
4. Select an analytic technique and justify the choice.
5. Design the dissemination product for a named consumer.
6. Write a feedback instrument and define what accuracy would be measured against.

Prerequisites

- Section 1.2 in full.
- Section 1.3 for discipline selection.
- Completion of Lab 1.
- Internet access is optional. If you choose Option A below, you will need it.

Instructions

Choose one scenario. Option A uses a real current situation and is harder. Option B is provided so the lab can be completed without internet access.

Option A (real). Select a decision facing a real organization that is publicly discussed: a company entering a market, a city considering an infrastructure investment, a university deciding on an enrollment strategy. You must be able to state the decision without insider information, and you may only use public sources. Do not select a scenario that centers on a private individual.

Option B (provided). **Northgate University**, a regional institution with 9,000 students, must decide by December whether to invest \$6M in a new online program division. Enrollment has been flat for three years. Two competing regional institutions have launched online programs in the past 18 months. State funding formulas are under legislative review.

Step 1 (20 min). Write the decision statement. One sentence. Name the decision maker, the decision, the resources at stake, and the deadline. Then write, in a second sentence, what would happen if the decision maker had perfect information.

Step 2 (25 min). Write one PIR and four to six EEIs. Then, for each EEI, write two to four indicators: specific observable things whose presence or absence helps answer it. Complete Worksheet 2A.

Step 3 (30 min). Build the collection plan on Worksheet 2B. For each indicator: which discipline, which specific source, what it would cost in time or money, what the legal or ethical considerations are, and how long collection would take. At least one row must be marked as a gap that cannot be collected.

Every plan has one, and pretending otherwise is the mistake this step is designed to catch.

Step 4 (20 min). Write the processing plan on Worksheet 2C. Identify at least four normalization or resolution problems your collected material would have. Consider: differing fiscal calendars, differing definitions of the same metric, name and entity variants, differing units or currencies, differing time zones, and duplicate reports of the same event.

Step 5 (30 min). Design the analysis. Select one structured analytic technique from Section 1.2.5 and justify why it fits this problem better than the alternatives. Then apply it at least partially: if you chose ACH, list your hypotheses and at least six pieces of evidence scored against them; if you chose a Key Assumptions Check, list at least six assumptions with the consequence of each being wrong.

Step 6 (25 min). Design dissemination. Complete Worksheet 2D: who is the consumer, what format, what length, what channel, what timing relative to the decision, what handling restrictions, and who must not receive it. Then write the first paragraph of the product as it would actually appear, in BLUF form.

Step 7 (20 min). Design feedback. Write four feedback questions for the consumer. Then define the accuracy test: state precisely what would have to be observed, and by when, for your assessment to be judged correct or incorrect. An assessment that cannot be scored later is not falsifiable, and this step exists to make you notice that.

Step 8 (10 min). Draw the lifecycle diagram for your scenario, showing the six phases with your specific content in each box, and draw at least one arrow that does not follow the clean circle: a place where a phase would loop back or jump forward in real practice. Explain that arrow in two sentences.

Worksheet 2A: Requirement Decomposition

Level	Content
Decision	
Decision maker	
Deadline	
PIR	
EEI 1	
Indicators for EEI 1	
EEI 2	
Indicators for EEI 2	
EEI 3	
Indicators for EEI 3	
EEI 4	
Indicators for EEI 4	

Worksheet 2B: Collection Plan

Indicator	Discipline	Specific source	Cost	Legal / ethical notes	Time to collect	Collectible?

Worksheet 2C: Processing Plan

Problem	Which sources are affected	Normalization or resolution step	What goes wrong if skipped

Worksheet 2D: Dissemination Design

Element	Your answer
Primary consumer	
Secondary consumers	
Who must NOT receive it	
Format and length	
Channel	
Timing relative to decision	
Handling restrictions	
First paragraph, BLUF form	

Expected Outcomes

- A PIR that is answerable, deadline-bound, and tied to the named decision. Requirements phrased as "tell me about X" score 1 or 0.
- At least twelve indicators total, and the majority should be genuinely observable rather than aspirational. "Their internal strategy documents" is not an indicator, because you cannot lawfully observe it.
- A collection plan that names specific sources, not categories. "Public records" scores 2; "state business registration database, searched monthly by entity name" scores 4.
- At least one honestly identified uncollectible gap.
- A processing plan that identifies at least one problem the student would plausibly have missed on a first pass, most often differing metric definitions across sources.
- A dissemination design where format matches consumer. A 20-page assessment for a board that

meets for 90 minutes is a wrong answer even if the content is excellent.

- An accuracy test that is falsifiable, with a date attached.

Discussion Questions

1. Which phase took you the longest? Which do you think would take longest in a real organization? Explain the difference.
2. Your collection plan has an uncollectible gap. How should the final product handle it: omit it, flag it, or reframe the assessment around it?
3. You chose one analytic technique. Name the technique you rejected and describe a version of this scenario where your rejected choice would have been correct.
4. Suppose the consumer reads your product and does nothing. List four distinct possible causes, one located in each of four different lifecycle phases.

Grading Rubric: Lab 2 (52 points)

Criterion	Weight	4 points looks like
Decision statement precision	x1	Names decision maker, resources, deadline, in one clear sentence
PIR quality	x2	Answerable, decision-linked, deadline-bound
EEl decomposition	x2	EEl collectively sufficient to answer the PIR, no overlap, no gaps
Indicator specificity	x2	Observable, discriminating, and realistically available
Collection plan realism	x3	Specific named sources, honest cost and timing, at least one honest gap
Legal / ethical annotation	x2	Each row assessed; anything questionable is flagged and avoided
Processing plan insight	x2	Four or more real problems identified with consequences
Analytic technique justification and application	x3	Correct technique for the problem, and actually applied, not merely named
Dissemination design fit	x2	Format, length, channel, and timing all match the named consumer

Criterion	Weight	4 points looks like
BLUF paragraph quality	x2	Judgment, confidence, and so-what in the opening
Feedback design and falsifiability	x2	Accuracy test is specific, observable, and dated
Lifecycle diagram with non-linear arrow	x1	Diagram accurate, and the real-world deviation is correctly explained

Total possible: 52 (13 criteria at listed weights, 4 points maximum each, weighted). Passing: 36 weighted points.

Lab 3: Identifying the Correct Intelligence Discipline

Time required: 60 to 90 minutes **Format:** Individual, then small group discussion **Difficulty:** Foundational to intermediate

Objectives

1. Match a stated intelligence requirement to the discipline or disciplines best suited to answering it.
2. Justify discipline selection on the basis of what each discipline can and cannot see.
3. Identify requirements that cannot be lawfully answered by a private sector actor.
4. Design a multi-discipline collection approach for a requirement that no single discipline can answer.

Prerequisites

- Section 1.3 in full, including the discipline comparison table in Section 2.4.
- The legal and ethical boundary table in Section 1.7.3.

Part 1: Discipline Matching (30 min)

For each of the eighteen requirements below, complete Worksheet 3A: identify the primary discipline, up to two supporting disciplines, and note whether a private sector actor could lawfully pursue it.

1. Determine whether a competitor's new factory is operational or still under construction.
2. Determine whether a competitor's CEO privately opposes the board's expansion plan.
3. Determine the bill of materials cost of a competitor's flagship product.
4. Determine which threat actor infrastructure is currently hosting phishing pages targeting our brand.

5. Determine whether a shell company received funds from a sanctioned entity.
6. Determine whether an underground facility is being used for chemical production.
7. Determine the current market rumor about an impending industry acquisition.
8. Determine the internal deliberations of a foreign ministry regarding a treaty.
9. Determine whether a supplier's factory was damaged by last week's flooding.
10. Determine how many people entered a competitor's store location per day last month.
11. Determine whether a specific employee is exfiltrating documents to a personal account.
12. Determine the acoustic signature of a vessel in a shipping lane.
13. Determine which regulatory changes are being drafted in a state legislature.
14. Determine whether a criminal group's leadership structure has changed after an arrest.
15. Determine the technical capability of a captured drone.
16. Determine whether a competitor's newly hired executives suggest a strategic shift.
17. Determine whether a person of interest was at a specific location at a specific time.
18. Determine whether public sentiment about our brand is shifting in a specific region.

Part 2: The Impossible Requirement (15 min)

Three of the eighteen requirements above cannot be lawfully pursued by a private company. Identify them on Worksheet 3B. For each, write:

- Which discipline would answer it.
- Which actor could lawfully pursue it, and under what authority.
- What lawful alternative requirement you would substitute, that gets the decision maker as close as possible to what they actually need.

That third item is the professional skill being taught here. A good analyst does not simply refuse an improper request. They reframe it into something lawful that still serves the decision.

Part 3: Multi-Discipline Design (30 min)

Take requirement 16 from Part 1: *determine whether a competitor's newly hired executives suggest a strategic shift.*

Design a multi-discipline collection approach on Worksheet 3C. Requirements:

- Use at least three disciplines.
- For each, state the specific collection activity and the specific finding it could produce.
- Identify which two disciplines, if they agreed, would raise your confidence most, and explain why. The reasoning should be about how hard it would be for the competitor to manipulate both

simultaneously, not about how much information each provides.

- Identify the one discipline you would drop if your budget were cut in half, and defend the choice.

Worksheet 3A: Discipline Matching

#	Primary discipline	Supporting disciplines	Lawful for a private company?	One-line justification
1				
...				
18				

Worksheet 3B: Reframing Improper Requirements

#	Discipline that would answer it	Who could lawfully do it	Lawful substitute requirement

Worksheet 3C: Multi-Discipline Design

Discipline	Collection activity	Possible finding	Cost tier (low/med/high)

Confidence pairing: which two disciplines agreeing would most raise confidence, and why?

Budget cut: which discipline do you drop, and what do you lose?

Expected Outcomes

- Requirements 2, 8, and 14 should be identified as HUMINT-primary, because each concerns intent, deliberation, or human structure rather than observable physical fact.
- Requirements 1, 9, and 10 should be identified as IMINT/GEOINT-primary. Requirement 10 in particular should be recognized as answerable through vehicle counting from commercial imagery, a well-established commercial practice.
- Requirement 3 is TECHINT via lawful retail purchase and teardown.
- Requirement 5 is FININT, and students should recognize that access is legally restricted.
- Requirements 4 and 11 are CYBINT, with the critical distinction that requirement 11 concerns an organization's own systems and employees, and is therefore lawful for the employer subject to policy and privacy law, whereas monitoring a person outside your authority is not.

- Requirement 6 is MASINT-primary, and requirement 12 is MASINT (acoustic).
- Requirements 7, 13, 16, and 18 are OSINT-primary.
- Requirement 15 is TECHINT, and students should note that "captured" implies a government context.
- The three requirements that private companies cannot lawfully pursue are **8** (foreign ministry internal deliberations, which would require SIGINT or recruited HUMINT against a foreign government), **12** (acoustic signature collection on a vessel, which requires sensors and authority a private firm does not have in that context), and **17** (locating a person of interest, which for a private actor edges into surveillance of an individual and generally requires either law enforcement authority or, in narrow cases, a licensed investigator operating within specific legal limits). Some students will nominate 5 or 6 instead, and a well-argued case for either is acceptable if the reasoning about authority is correct.
- The multi-discipline design should reach a confidence pairing argument based on independence and manipulation cost, not volume.

Discussion Questions

1. Several requirements could be answered by more than one discipline. When is redundant collection worth its cost, and when is it waste?
2. OSINT was the primary discipline for four requirements and a supporting discipline for many more. Does that make it the most important discipline, or the cheapest? Defend your answer.
3. In Part 2, you reframed an improper requirement. Did your substitute actually serve the decision maker's need, or did it quietly answer a different question? Be honest.
4. Requirement 11 involves monitoring an employee. What policies would need to exist before you would consider that lawful and ethical, and what would you refuse to do even if it were lawful?

Grading Rubric: Lab 3 (36 points)

Criterion	Weight	4 points looks like
Discipline matching accuracy (Part 1)	x3	15 or more of 18 correct with sound justification
Justification quality	x2	Reasoning refers to what the discipline can and cannot see, not to keywords in the prompt
Legal / lawfulness assessment	x2	Correct calls, and the reasoning cites authority rather than intuition
Improper requirement identification (Part 2)	x1	Correct three, or a defensible alternative set with correct reasoning

Criterion	Weight	4 points looks like
Quality of lawful substitutes	x2	Substitutes actually serve the underlying decision
Multi-discipline design (Part 3)	x2	Three or more disciplines with specific activities and plausible findings
Confidence pairing reasoning	x2	Argument based on source independence and manipulation cost
Budget-cut defense	x1	Names a real trade-off and accepts the loss explicitly

Total possible: 36 weighted. Passing: 25.

Lab 4: Strategic, Tactical, and Operational Decisions in One Organization

Time required: 2 to 2.5 hours **Format:** Small groups of three, then individual write-up **Difficulty:** Intermediate

Objectives

1. Classify decisions by level and defend classifications at the boundaries.
2. Produce three intelligence products at three different levels from the same underlying situation.
3. Demonstrate how the levels feed each other in both directions.
4. Diagnose the organizational symptom produced by a missing level.

Prerequisites

- Sections 1.4, 1.5, and 1.6.
- The three-level comparison table in Section 2.2.
- Completion of Lab 1.

The Fictional Organization

Ironvale Utilities is a regional water and electric utility serving 380,000 customers across a six-county area.

- 1,400 employees, including a 12-person security team and one intelligence analyst.
- Infrastructure: 3 generation facilities, 2 water treatment plants, 210 substations, roughly 4,000 miles of distribution.
- Regulatory environment: state utility commission oversight, plus federal reliability standards.

- Recent history: two ransomware attempts against peer utilities in adjacent states in the past year; a physical intrusion at one of Ironvale's own substations six weeks ago (fence cut, no equipment damage, no arrest); a state legislature currently considering a bill that would require utilities to fund a regional resilience program; aging infrastructure with a 20-year replacement backlog and a rate case pending.

Part 1: Decision Classification (30 min, group)

Classify each of the following fourteen decisions as strategic, operational, or tactical on Worksheet 4A, with a one-line justification. Four of them are deliberately near the boundary between two levels.

Identify which four you found hardest and why.

1. Whether to isolate a specific workstation showing suspicious authentication activity, right now.
2. Whether to invest \$40M over eight years in grid segmentation and physical hardening.
3. How to sequence a six-month program to reduce substation intrusion across the highest-risk 30 sites.
4. Whether to dispatch a patrol to a specific substation tonight based on a reported vehicle.
5. Whether to support or oppose the pending resilience legislation.
6. Which detection rules to deploy this week based on indicators shared by a peer utility.
7. Whether to reorganize the security team to add a dedicated threat hunting function.
8. Whether to hire a second intelligence analyst.
9. How to allocate the security team's field time across six counties for the next quarter.
10. Whether to notify the state commission about the substation intrusion.
11. Whether the company's 20-year infrastructure plan should assume increasing or stable threat levels.
12. Which vendor remote-access accounts to disable immediately following a vendor breach announcement.
13. How to structure the vendor risk program so that a vendor breach does not require an emergency response next time.
14. Whether to open a joint operations arrangement with the regional fusion center.

Part 2: Three Products, One Situation (60 min, group with individual sections)

The situation: **the substation fence-cutting intrusion six weeks ago, plus two similar reported intrusions at other utilities in the region in the same period.**

Each group member writes one of the following. If working alone, write all three.

Product A: Tactical bulletin. One paragraph, for field crews and patrol. What to look for, what to do if seen, who to call. Maximum 120 words.

Product B: Operational assessment. One to two pages, for the security director. Describe the pattern across the three incidents, assess what the activity is likely preparation for, identify the 30 highest-risk sites and the criteria you used to rank them, map where detection is weakest, and recommend a six-month program with sequencing.

Product C: Strategic assessment. One to two pages, for the CEO and board, timed to the capital planning cycle. Assess whether physical threat to distribution infrastructure is likely to increase over the next five to ten years, what that means for the \$40M hardening decision and the 20-year replacement backlog, and how the pending legislation changes the calculation. Present at least two scenarios with indicators that would tell leadership which one is developing.

Each product must carry a judgment and a confidence level. Each must name at least one gap.

Part 3: The Feeds (20 min, group)

On Worksheet 4B, map the flows:

- What does the strategic product need *from* the operational level to be credible?
- What does the operational product need *from* the tactical level?
- What does the tactical level need *from* the operational level to avoid being a treadmill?
- What does the strategic level *give* to the operational level?

Then answer: Ironvale has one analyst. Which level will actually get produced, and which will silently not get produced? What symptom will the organization display in two years as a result?

Part 4: Individual Write-Up (30 min)

Write a one-page memo to Ironvale's CEO arguing for or against hiring a second intelligence analyst. Use your Part 3 findings. This is an argument, not an assessment, so you may advocate. But you must state honestly what the strongest counterargument is and answer it.

Worksheet 4A: Decision Classification

#	Level	Justification	Boundary case?
1			
...			
14			

Worksheet 4B: Level Interdependence

Flow	What is passed	What breaks if the flow is absent
Tactical to operational		
Operational to strategic		

Flow	What is passed	What breaks if the flow is absent
Strategic to operational		
Operational to tactical		

Expected Outcomes

- Clear tactical: 1, 4, 6, 12. Clear operational: 3, 9, 13. Clear strategic: 2, 5, 11.
- Boundary cases most groups will struggle with: 7 (reorganizing the team is arguably strategic in intent and operational in execution), 8 (hiring one person is a small resource decision with strategic implications for capability), 10 (notification is immediate, which feels tactical, but it is a governance and regulatory posture decision), and 14 (a partnership is operational in function and strategic in commitment). Credit is awarded for the quality of the argument, not for matching a single answer key.
- Product A should be under 120 words, contain no analysis, and be immediately usable by someone in a truck.
- Product B should rank sites using stated criteria, which is the step students most often skip. A ranking with no stated criteria scores 2 at best.
- Product C should present scenarios rather than a single prediction, and should include indicators that discriminate between scenarios.
- Part 3 should conclude that the operational layer is the one that will not get produced, because tactical work is urgent and strategic work is executive-facing. The predicted symptom is a security function that responds competently to individual incidents indefinitely without reducing their frequency.

Discussion Questions

1. Compare Products A and C. Same underlying facts. What percentage of the content is shared? What does that tell you about the claim that intelligence products are "just the facts"?
2. Product B required you to rank 30 sites. Where did your ranking criteria come from, and are they defensible if a site you ranked 29th is attacked?
3. In Part 1, four decisions were near a boundary. Does the ambiguity mean the three-level model is flawed, or does it mean the model is doing work by forcing the question?
4. If Ironvale's board reads only Product C and never sees B, what specific risk does that create for the credibility of the strategic assessment?

Grading Rubric: Lab 4 (48 points)

Criterion	Weight	4 points looks like
Classification accuracy on clear cases	x2	All ten clear-cut decisions correct
Boundary case reasoning	x2	Ambiguity recognized and argued, not guessed
Product A: tactical discipline	x2	Under 120 words, zero analysis, immediately actionable
Product B: operational structure	x3	Pattern assessed, ranking criteria stated, detection gaps mapped, program sequenced
Product C: strategic scenarios	x3	Two or more scenarios with discriminating indicators, tied to the capital decision
Confidence levels across all three products	x2	Present, defensible, and appropriately different across levels
Level interdependence mapping	x2	Both directions correct, with consequences of absence
Diagnosis of the missing layer	x2	Correctly identifies operational, and names a realistic two-year symptom
Individual memo argument quality	x2	Takes a position, states the strongest counterargument, answers it

Total possible: 48 weighted. Passing: 34.

Lab 5: Analyzing a Recent Public Event Through Multiple Disciplines

Time required: 2.5 to 3.5 hours **Format:** Individual **Difficulty:** Advanced

Objectives

1. Apply the full discipline framework to an unstructured real-world event.
2. Distinguish what is known from what is being asserted in public reporting.
3. Identify how each discipline could contribute to understanding, including disciplines you cannot personally use.
4. Recognize circular reporting in live open source material.

5. Produce a professional assessment about a real situation with honest confidence calibration.

Prerequisites

- All of Part 1 of the reading material.
- Completion of Labs 1 through 4.
- Internet access.

Event Selection Rules

Read these carefully. They are graded.

Choose an event that is:

- Public, recent (within the last 12 months), and covered by multiple independent outlets.
- Institutional or systemic in nature: a supply chain disruption, an infrastructure failure, a major product launch or market entry, a natural disaster response, a regulatory action, a corporate restructuring, a large-scale cyber incident, an election administration issue, a public health event.

Do not choose:

- Anything centered on a private individual. Not a crime victim, not a suspect, not a person who became briefly famous. This is a firm rule. Intelligence methods aimed at private individuals is a different activity with different rules, and this lab does not authorize it.
- An ongoing criminal case where your analysis could be mistaken for an accusation.
- An event about which you hold a strong prior political commitment, unless you are prepared to have that commitment tested and to document how you tested it. Choosing such an event is permitted and is genuinely valuable, but Step 6 will be harder and it is graded harder.

Step-by-Step Instructions

Step 1 (15 min). State the event in three sentences: what happened, when, and why it matters to some decision maker. Name the decision maker whose perspective you will adopt for the rest of the lab. This choice shapes everything: an insurer, a regulator, a competitor, a city manager, and a supply chain director all need different intelligence about the same event.

Step 2 (30 min). Build a fact inventory on Worksheet 5A. Fifteen or more items. For each, record the claim, the outlet, whether the outlet appears to have independent reporting or is citing another outlet, the date, and your reliability/credibility rating. Do not include anything you cannot attribute.

Step 3 (25 min). Circular reporting audit. On Worksheet 5B, trace at least three claims back toward their original source. For each, record how many outlets carried it, how many appear to have reported it independently, and what the earliest identifiable source is. You are looking for the pattern where eight outlets create the impression of corroboration and all eight cite one original report. Note at least one claim where you could not determine the original source, and state how that affects your confidence.

Step 4 (35 min). Discipline contribution analysis on Worksheet 5C. For each of the eight disciplines, answer:

- What could this discipline contribute to understanding this event?
- Who would have the access and authority to use it here?
- Could you personally use it lawfully? If not, say so plainly.
- What specific question would it answer that no other discipline could?

Some disciplines will not apply. Saying "MASINT contributes nothing here, because there is no physical signature at issue" is a correct and complete answer, and it earns full credit. Manufacturing a use for every discipline is a failure mode this step is designed to expose.

Step 5 (25 min). Gap analysis. List the eight to ten most important things that are still unknown about this event publicly. For each: which discipline would resolve it, who could collect it, and how the answer would change the picture. Rank the gaps by how much resolving them would change your assessment. That ranking is the most valuable output of this step.

Step 6 (25 min). Bias check. Complete Worksheet 5D:

- Write your initial belief about this event, as it was before Step 2.
- List three pieces of information you encountered that contradicted it.
- State honestly how you handled each: did you weight it, dismiss it, or look for a reason to discount it?
- Write the strongest version of the explanation you find least plausible. Not a caricature. The strongest version, argued in good faith, in at least four sentences.
- State what evidence would move you.

Step 7 (35 min). Write the assessment. Two pages maximum, using the Section 1.10 template, addressed to the decision maker you named in Step 1. It must contain:

- A BLUF judgment with a confidence level.
- Facts clearly separated from assessment.
- At least two competing explanations, with your reasoning for preferring one.
- Key assumptions with consequences.
- Ranked intelligence gaps.
- Recommendations appropriate to that specific decision maker.
- A short section titled "What would change this assessment."

Step 8 (10 min). Reliability appendix. One paragraph: state where your weakest sourcing is, and what a reader should discount as a result. Analysts who write this section honestly are more trusted over time, not less.

Worksheet 5A: Fact Inventory

#	Claim	Outlet	Independent or citing?	Date	Rel.	Cred.
1						
...						

Worksheet 5B: Circular Reporting Audit

Claim	Outlets carrying it	Apparently independent	Earliest identifiable source	Effect on confidence

Worksheet 5C: Discipline Contribution

Discipline	Potential contribution	Who has the access	Lawful for me?	Unique question it answers
OSINT				
HUMINT				
SIGINT				
IMINT / GEOINT				
MASINT				
TECHINT				
FININT				
CYBINT / CTI				

Worksheet 5D: Bias Check

Prompt	Your answer
My initial belief was	
Contradicting information I found	
How I actually handled it	
Strongest version of the view I find least plausible	
Evidence that would change my mind	

Expected Outcomes

- An event that satisfies the selection rules. Choosing an event centered on a private individual results in a zero for the lab regardless of analytic quality, and the reason is stated in the objectives.
- A fact inventory where at least a third of the items are traced to citing rather than independent reporting. This is the normal state of public reporting on recent events, and discovering it is one of the main learning outcomes.
- At least one claim where the original source could not be determined, honestly reported.
- A discipline analysis that plainly states which disciplines the student cannot lawfully use, with at least two disciplines correctly identified as inapplicable.
- Confidence calibration that is *lower* than the student's initial instinct. Students who complete Steps 3 and 6 honestly almost always downgrade. A submission that reaches high confidence on a recent, still-developing event should be scrutinized closely and usually indicates the bias check was performed superficially.
- A steel-manned opposing view in Step 6 that a person holding that view would recognize as fair. This is the single hardest criterion in the lab and the most valuable.

Discussion Questions

1. How much of what you initially "knew" about this event survived the sourcing audit intact?
2. Your confidence probably went down as you did more work. Explain to a skeptical executive why that is a sign of good analysis rather than of a bad analyst.
3. Which discipline, unavailable to you, would have contributed most? What does that tell you about the ceiling on open source analysis?
4. Compare your Step 6 steel-manned argument to your actual assessment. Did writing the opposing case change anything in your final product? If it changed nothing at all, consider whether you wrote it before or after you finished the assessment.

Grading Rubric: Lab 5 (60 points)

Criterion	Weight	4 points looks like
Event selection compliance	x1	Meets all rules; institutional rather than personal
Decision maker framing	x2	A specific consumer named and consistently served throughout
Fact inventory quality	x2	15+ items, all attributed, ratings defensible
Independence assessment	x3	Citing versus independent correctly distinguished across most items

Criterion	Weight	4 points looks like
Circular reporting audit	x3	Three or more claims traced; at least one unresolved and honestly reported
Discipline contribution analysis	x3	All eight addressed; inapplicable ones correctly dismissed; unique questions identified
Lawfulness self-assessment	x2	Clear and correct about what the student cannot do
Gap analysis and ranking	x3	Gaps specific, disciplines matched, ranking justified by analytic impact
Bias check honesty	x3	Contradicting evidence engaged rather than dismissed; handling described candidly
Steel-manning quality	x3	Opposing view stated in a form its holders would accept as fair
Assessment structure and BLUF	x2	Template followed, judgment first, facts separated from assessment
Confidence calibration	x3	Level defensible given the sourcing; overconfidence penalized
Reliability appendix	x1	Weakest sourcing named without hedging

Total possible: 60 weighted. Passing: 42.

Part 9: Capstone Assignment

The Meridian Port Authority Scenario

Weight: 30% of chapter grade **Time required:** 4 to 6 hours **Format:** Individual. May be completed over multiple sittings. **Deliverables:** Five items, listed at the end.

9.1 The Scenario

You have been hired as the first intelligence analyst for the **Meridian Port Authority (MPA)**, a mid-sized inland port and intermodal freight hub. This is a fictional organization and a fictional scenario. Everything you need is contained below.

About MPA:

- Operates a river port, a rail intermodal yard, and 1.2 million square feet of leased warehousing.
- 620 employees. Roughly 40 tenant companies operate on MPA property.
- Governed by a seven-member appointed board. Reports to a state transportation authority.
- Revenue: lease income, tonnage fees, rail service fees. Annual operating budget approximately \$95M.
- Security posture: a 22-person security department focused on physical access and theft. A 6-person IT team. No prior intelligence function. No threat intelligence subscriptions.
- Current capital question: the board is deciding whether to commit \$52M over six years to an automation and terminal modernization program. A decision is expected in five months.

Your position: you report to the Director of Security, who reports to the Chief Operating Officer. The board has asked for "better situational awareness" without specifying what that means. Translating that into something answerable is part of your job and part of this assignment.

9.2 The Raw Reporting

You have been handed the following 24 items, collected over the past 90 days from a mix of internal records, public sources, partner organizations, and conversations. They arrived unsorted and unevaluated. Some are important. Some are noise. At least two are contradictory. At least one is almost certainly false.

#	Item	Provenance
1	Cargo theft reports on MPA property rose from 4 to 14 incidents in the last two quarters.	MPA internal security records
2	Of those 14, 11 involved high-value electronics from a single tenant's warehouse.	MPA internal security records
3	In 9 of the 14, the stolen containers were among the last three received before the theft, suggesting the thieves knew which containers held what.	MPA internal security analysis
4	The tenant in question uses a third-party logistics provider whose driver check-in system is web-based and accessible from the public internet.	Observed by MPA IT during an unrelated review
5	A regional trucking association	Partner organization, association

#	Item	Provenance
	bulletin describes an organized cargo theft group operating along the corridor, using fraudulent pickup documentation.	has been accurate in past bulletins
6	An anonymous tip to MPA's ethics line alleges that "someone in the gate office is selling manifest information."	Anonymous, uncorroborated
7	MPA gate office staffing was reduced by three positions eight months ago as a cost measure; overtime in that office is up 40%.	MPA HR and finance records
8	Two peer inland ports in adjacent states disclosed ransomware incidents in the past 14 months. One publicly reported a three-week disruption to gate operations.	Public disclosures and trade press
9	MPA's terminal operating system runs on a version the vendor will stop supporting in 11 months.	MPA IT records
10	A vendor that provides MPA's crane maintenance announced a breach of its remote support platform 20 days ago. MPA has an active remote support account with them.	Vendor notification letter
11	Rail volume through MPA is up 7% year over year.	MPA operational data
12	River water levels are forecast below the 20-year average for the coming season, per a federal forecast.	Public federal forecast
13	A competing port 140 miles away announced a \$200M expansion, including automation, with completion projected in four years.	Public announcement and trade press
14	Two of MPA's largest tenants have leases expiring within 18 months. Combined, they represent 21% of lease revenue.	MPA lease records

#	Item	Provenance
15	One of those two tenants requested a site tour of the competing port, according to a broker who works with both.	Broker conversation; broker is a repeat contact, generally reliable, but has an incentive to create urgency
16	A social media post claims MPA "is about to lay off half the gate staff." No source. Post has 40 shares.	Public social media, anonymous
17	MPA's insurance carrier has requested documentation of the port's cyber controls before renewal in four months.	Carrier correspondence
18	State legislature is considering a bill that would fund inland port infrastructure grants, currently in committee.	Public legislative tracking
19	Local news reported a fence breach at MPA's north perimeter three weeks ago. No arrest, nothing taken.	Local news and MPA records, both consistent
20	A drone was observed over the intermodal yard twice in the past month by security staff. No identification.	MPA security incident reports, firsthand
21	Job postings by the competing port include four automation engineer roles posted in the last 60 days.	Public job board
22	MPA's own automation feasibility study, commissioned last year, concluded that modernization would reduce gate processing time by 30%.	Internal consultant report
23	An MPA gate supervisor told a colleague that "everybody knows the check-in system password."	Secondhand, single source, no verification attempted
24	Fuel prices region-wide rose 9% this quarter.	Trade publication

9.3 What You Must Produce

Deliverable 1: Classification Worksheet (15 points)

For all 24 items, produce a table with:

- Ladder position: data, information, or knowledge.
- Reliability rating (A through F) and credibility rating (1 through 6), each with a one-line justification.
- Relevance: relevant, possibly relevant, or noise, judged against the requirements you develop in Deliverable 2.
- The discipline or source category the item came from.

Then, in one paragraph, identify the two items that most directly contradict each other and explain how you resolved the contradiction. Also identify the item you believe is almost certainly false and state why.

Deliverable 2: Requirements Package (20 points)

The board asked for "better situational awareness." That is not a requirement. Convert it.

Produce:

- **Three PIRs**, each tied to a named decision the board or executive team actually faces. At least one must support the \$52M modernization decision.
- **For each PIR: three to five EEIs.**
- **For each EEI: two to four indicators**, each observable.
- A prioritization statement: which PIR is first, and why. Resources are finite. You are one person.

You will be graded on whether your PIRs are answerable and decision-linked, not on how comprehensive they sound.

Deliverable 3: Lifecycle Plan (20 points)

Map your highest-priority PIR through all six phases. For each phase, produce the specific content, not a description of the phase:

- **Planning:** the PIR, EEIs, indicators, deadline, and named decision maker.
- **Collection:** a plan table with each indicator assigned to a discipline and a specific source, with cost, time, and legal or ethical notes. Include at least one gap you cannot collect, and say so.
- **Processing:** at least four specific normalization or entity resolution problems your material will have, with the consequence of skipping each.
- **Analysis:** name and apply one structured analytic technique. If you choose ACH, provide the

matrix with at least three hypotheses and eight pieces of evidence. If you choose a Key Assumptions Check, list at least six assumptions with consequences.

- **Dissemination:** consumer, format, length, channel, timing, handling, and who must not receive it.
- **Feedback:** four feedback questions plus a falsifiable accuracy test with a date.

Deliverable 4: Discipline Assessment (15 points)

A table covering all eight disciplines. For each:

- What it could contribute to MPA's requirements specifically.
- Whether MPA can lawfully and practically use it. Be specific about why not, where applicable.
- Priority for MPA: build now, build later, or not applicable.
- If "build now," what it would cost approximately, in people or dollars, and what it would replace in your workload.

Then, in one paragraph, recommend which two disciplines MPA should invest in first and defend the choice on the basis of what MPA's actual decisions require, not on general merit.

Deliverable 5: Professional Intelligence Summary (30 points)

The main product. **Maximum two pages.** Addressed to the Chief Operating Officer, timed to reach the board before the modernization decision.

Required structure:

1. **Bottom line.** No more than four sentences. A judgment on the most important thing MPA faces, with a confidence level and estimative probability language where you forecast.
2. **Key facts.** Six to ten, each with its reliability/credibility rating in brackets. Observed facts only, no assessment.
3. **Assessment.** What the facts mean. Must include at least one alternative explanation you considered and rejected, with your reasoning for rejecting it.
4. **Key assumptions.** At least three, each with a statement of what changes if it is wrong.
5. **Intelligence gaps.** At least four, ranked by how much resolving them would change your assessment.
6. **Recommended actions.** At least five, each specific, assignable to a named role, and time-bound. At least one must be something MPA can do this week at no cost.
7. **What would change this assessment.** Two to four sentences.
8. **Collection requests.** What you will task next.

Below the summary, on a separate page, attach a **rationale annex** of no more than one page explaining:

- Why you selected this judgment as the most important thing, out of everything in the reporting.
 - Why you chose your confidence level and not the level above or below it.
 - Which two items you excluded that a less careful analyst would have included, and why.
-

9.4 Guidance on the Substance

A note that is not a hint about the answer, but is a hint about the shape of a good answer.

A weak submission treats the 24 items as 24 separate problems and produces a list of concerns. A strong submission notices that several items describe **one connected exposure** and several others describe **an unrelated but genuinely important strategic question**, and it makes a choice about which one goes in the bottom line. Intelligence requires that choice. A product that says "here are eleven things to worry about" has transferred the analytic burden back to the reader, which is exactly the work the reader was paying you to do.

You will also find that some items look alarming and are not, and at least one item that looks mundane is the most important thing in the set. That inversion is deliberate.

9.5 Capstone Grading Rubric (100 points)

Deliverable	Criterion	Points
1. Classification	Ladder classification accuracy across 24 items	4
	Reliability and credibility ratings defensible and consistent	4
	Noise correctly identified	3
	Contradiction identified and resolution reasoned	2
	Probable falsehood correctly identified with reasoning	2
2. Requirements	Three PIRs, each decision-linked and answerable	8
	EEIs collectively sufficient, non-overlapping	5
	Indicators observable and discriminating	4
	Prioritization defended against a resource constraint	3
3. Lifecycle	Planning content specific and complete	3

Deliverable	Criterion	Points
	Collection plan names specific sources, includes an honest gap	5
	Legal and ethical notes present and correct	3
	Processing problems real, with consequences	3
	Analytic technique correctly applied, not merely named	4
	Dissemination design fits the consumer	2
4. Disciplines	All eight addressed with MPA-specific reasoning	6
	Lawfulness and practicality judgments correct	4
	Investment recommendation tied to MPA's decisions	5
5. Summary	BLUF: judgment, confidence, estimative language, four sentences or fewer	5
	Correct selection of the most important judgment	5
	Facts separated from assessment, all rated	4
	Alternative explanation considered and reasoned	4
	Assumptions with consequences	3
	Gaps ranked by analytic impact	3
	Recommendations specific, assignable, time-bound, including a no-cost immediate action	4
	Length discipline (two pages)	2
	Rationale annex: confidence defended against adjacent levels	4
	Rationale annex: exclusions identified and justified	3
Overall	Professional presentation, internal consistency, no unsupported claims	3

Deliverable	Criterion	Points
Total		100

Grade bands: 90 to 100 exemplary, 80 to 89 proficient, 70 to 79 competent, 60 to 69 developing, below 60 not yet passing.

Automatic deductions:

- Any judgment stated without a confidence level: minus 5 per instance.
- No intelligence gaps named anywhere in the submission: minus 10.
- Summary exceeding two pages: minus 5.
- Any recommendation not assignable to a role: minus 2 per instance.
- Presenting an assessment as a fact: minus 5 per instance.

Part 10: Answer Keys

Instructors may wish to remove this section from student copies of the workbook.

10.1 Short Answer Key

1. Information is organized fact that exists independently of need. Intelligence is information collected against a requirement, evaluated, analyzed in context, judged with a confidence level, and delivered to a specific consumer in time to affect a decision. The five required elements are purpose, evaluation, analysis, timeliness, and an explicit judgment.

2. Collection capacity and analytic capacity are separate. Added collection increases the volume of material an unchanged analytic capacity must filter, which reduces the proportion of collected material that ever gets examined. Example: a department adding 200 cameras and no analysts produces thousands of daily camera hours and the same two analysts, so usable intelligence output barely changes while storage cost and false confidence both rise.

3. Data is raw values with no meaning. Information adds context, labeling, and organization so a human can understand what is described. Knowledge adds comparison against history and baselines so patterns emerge. Intelligence adds judgment, forecast, confidence, and accountability.

4. Reliability judges the provider's track record. Credibility judges whether this specific report is plausible on its own merits. They must be separate because a reliable source can pass along something false and an unrated source can report something true; collapsing them causes analysts either to accept errors from trusted sources or discard accurate reporting from new ones.

5. B2 means a usually reliable source reporting something probably true and consistent with other reporting. F3 means a source that cannot be evaluated reporting something plausible but uncorroborated. B2 can support a judgment; F3 should be caveated and generally cannot carry a

judgment alone.

6. Planning and direction defines the requirement. Collection gathers raw information against the plan. Processing and exploitation makes collected material usable and comparable. Analysis and production turns evaluated information into a judged product. Dissemination delivers it to the right consumer in a usable form and time. Feedback and evaluation assesses usefulness and accuracy and generates new requirements.

7. A decision is what the consumer must choose. A PIR is the endorsed question whose answer would change that choice. EEIs are the sub-questions that must be answered to satisfy the PIR. Indicators are the observable things that answer EEIs. Example: decision to fund a distribution center; PIR on competitor entry within 18 months; EEI on regional warehouse capacity; indicator being a commercial lease filing in that metro.

8. It names a decision; it can be answered yes, no, or probably; it has a deadline tied to the decision; and answering it would change what the decision maker does.

9. Processing produces no visible product, so it receives little attention, but every downstream judgment depends on it. Two failures: mixed time zones across sources produce a false event sequence, causing analysts to describe a causal chain that never happened; failed entity resolution either splits one actor into three, hiding a pattern, or merges three into one, inventing a relationship.

10. Confidence describes how solid the basis for a judgment is, given sourcing, corroboration, and assumption load. Probability describes how likely the event is. Correct combined use: "We assess with high confidence that an attack in the next quarter is unlikely," meaning the analyst is very sure the odds are low.

11. Analysis of Competing Hypotheses counters confirmation bias by forcing disconfirmation across alternatives. Key Assumptions Check counters unexamined premises and layering. Devil's advocacy or red teaming counters groupthink and mirror imaging. Indicators and warnings counters late recognition of change.

12. It is skipped because it is never urgent and the next requirement is always waiting. The cost is that the function cannot distinguish products that changed a decision from products that were filed unread, so it cannot improve, and it will keep producing both indefinitely.

13. BLUF places the judgment in the first sentence, with support following. Academic structure builds to a conclusion at the end, which fails because intelligence consumers frequently stop reading after the first paragraph and must be able to act correctly on that paragraph alone.

14. A discipline is a category of collection defined by source type and method. Counterterrorism is a subject or mission, not a source; it is pursued using HUMINT, SIGINT, OSINT, FININT, and others, which is why it cannot be a discipline.

15. OSINT is cheap, fast, lawful, and broadly shareable, but is high in volume, vulnerable to deception, and prone to circular reporting. HUMINT is slow, expensive, and carries the highest legal and ethical risk, but uniquely reaches intent, internal deliberation, plans not yet acted on, and disagreement. HUMINT is worth its cost when the decision depends on intent rather than on observable capability,

and when the cost of being wrong about intent is high.

16. Strategic: one to ten-plus years, senior leadership, scenario and assessment products. Operational: weeks to a year, campaign and case managers, network charts and campaign assessments. Tactical: seconds to days, operators, bulletins and indicator products.

17. Tactical work is urgent so it always gets done; strategic work is executive-facing so it gets funded; operational work is neither, so it is deferred. The symptom is an organization that is continuously busy, executes individual responses competently, and never reduces the frequency of the events it responds to.

18. The identity rule: if you would have to misrepresent who you are or who you work for, do not collect it. The publication rule: if the owner took reasonable measures to keep the information secret and you are working to defeat those measures, you are outside the profession. The first prevents deceptive elicitation and pretexting; the second tracks trade secret law's protection of information the owner actively safeguarded.

19. Business intelligence looks inward at the organization's own data and answers what happened and why. Competitive intelligence looks outward at external actors and answers what others will do. Excellent BI with no CI produces an organization that is precisely informed about its own past and blind to an external change, because internal data records the consequences of competitor action only after it occurs.

20. The tactical indicators were already available and unused; the same alert had fired 60 times and been closed as a false positive. What changed the outcome was operational intelligence, which established the actor's technique sequence and told the analyst that authentication to backup infrastructure meant encryption was imminent, and strategic intelligence, which had made this threat class a standing priority worth writing a requirement against.

10.2 Multiple Choice Key

#	Answer	Note
1	b	Judgment plus confidence is the defining addition
2	c	Collection capacity rose, analytic capacity did not
3	b	Reliability is about track record; credibility about this report
4	c	An ill-defined requirement wastes the whole cycle
5	c	A specific observable sub-question
6	b	Normalization is processing
7	b	Confidence and probability are different measures

#	Answer	Note
8	c	Only HUMINT reliably reaches intent
9	b	Lawful teardown is standard TECHINT practice
10	b	Apparent corroboration from a single origin
11	c	Perishable, specific, operator-facing
12	b	Structure and choke points are operational
13	c	Misrepresentation of identity is the disqualifying element
14	b	Assuming the adversary shares your reasoning
15	c	Pre-deciding moved the decision earlier

10.3 True or False Key

#	Answer	If false, why
1	False	Intelligence adds requirement-driven collection, evaluation, analysis, judgment, confidence, and timely delivery to a specific consumer
2	False	Timeliness is a defining element; an analysis that arrives after the decision is history, not intelligence
3	False	Reliability judges the source's track record; credibility judges the plausibility of the specific report
4	False	In practice phases overlap, run in parallel, and sometimes jump forward or loop back
5	True	
6	True	
7	False	The distinction is collection

#	Answer	If false, why
		method and lawfulness, not analytic method; the analysis is often identical
8	False	Reversed: BI looks inward at internal data, CI looks outward at external actors
9	True	
10	True	

10.4 Critical Thinking Exercise Guidance

7.1 Strategic. Multi-year horizon, board-level consumer, threat actor categories rather than specific events, tied to a capability investment decision. Disciplines: strategic CTI, OSINT, sector sharing.

7.2 Tactical. Seconds to minutes, operator consumer, a specific indicator with a specific instruction, pushed rather than pulled. Discipline: tactical CTI plus owned telemetry.

7.3 Operational. Structure and choke points across a network, a multi-month case plan, supervisor-level consumer. Disciplines: records analysis, link and network analysis, FININT for the cash business.

7.4 Business intelligence, descriptive to diagnostic, and *not* intelligence in the analytic sense as described. It reports internal past performance with no external judgment, no forecast, and no confidence level. Students should notice that this is the trap in the set.

7.5 Competitive intelligence at the operational level, using GEOINT via commercial imagery. Lawful, because the imagery is commercially available and the lots are publicly visible. Note that this is a well-established commercial practice.

7.6 Strategic analysis producing an operational trigger with tactical execution. The matrix itself is an indicators and warnings product. The key insight students should reach: its value is that it pre-decides.

7.7 The problem is format and language, not content. A tactical consumer 20 minutes from entry cannot act on "indeterminate pending further collection." Rewrite: "Armament unconfirmed. Subject has two prior weapons arrests. Treat as armed." Uncertainty is preserved but converted into an actionable default.

7.8 This is a data feed, not intelligence. It has no requirement behind it, no evaluation, no analysis of relevance to this environment, no judgment, and no recommendation. It may be a useful input to a CTI program. Sold as complete intelligence, the claim is false. Good answers will note that 90,000 unfiltered indicators can actively harm a small team by generating unmanageable alert volume.

7.9 Confirmation bias, with an element of undocumented dismissal that will later look like layering. Prescribe Analysis of Competing Hypotheses, which forces the routine-exercise hypothesis to be scored explicitly against every piece of evidence rather than set aside. A Key Assumptions Check is also acceptable.

7.10 Stop. The document is another company's confidential material, and using it exposes the company to trade secret liability regardless of analytic quality. Correct handling: quarantine the document, do not distribute it, notify legal, remove any findings derived from it, and document the removal. Excellence of the analysis is aggravating rather than mitigating, because it demonstrates the value derived. Students should also note this is a personnel and onboarding failure, not only an analyst's problem.

7.11 Layering. A low-confidence judgment has been silently promoted to established background fact across successive products, and the uncertainty has disappeared without anyone deciding to remove it. Remedy: carry confidence forward with the claim, and cite the original product.

7.12 Any well-formed rewrite should name a decision, be answerable, be deadline-bound, and be tied to a change in action. Example: "Which ransomware operators have targeted organizations of our size in our sector in the past 12 months, which initial access methods did they use, and which of those methods are currently viable against our environment? Needed before the security budget submission in six weeks." Better because it is answerable, scoped, bounded by our own environment, and tied to a decision with a date.

7.13 Technical concern: the model is trained on arrest data, which measures police activity as much as criminal activity, producing a feedback loop where historically over-policed areas generate more arrests, which directs more patrol, which generates more arrests. Civil rights concern: disparate impact on specific communities, with the appearance of objectivity supplied by the model. Safeguards may include auditing input data for known enforcement bias, using victim reports rather than arrests where possible, measuring outcomes by community, holding a named human accountable for every allocation decision, publishing the methodology, and setting review intervals.

7.14 Diagnosis: the organization has collection and processing capacity, a semantic and consumption layer, and no analysis layer. It produces information at scale and no intelligence. Vocabulary to use: dashboard proliferation, vanity metrics, missing so-what, descriptive-only maturity. What to add: analysts with the mandate to write judgments and recommendations, requirements tied to named decisions, and a feedback mechanism asking whether anything changed.

7.15 Not usable. "Meaningful probability" has no defined value, so different readers will interpret it differently. "Sustained rainfall exceeding historical norms" defines no threshold, no duration, and no measurable trigger. There is no confidence level, no time frame, no consequence description, and no recommended action or decision authority. A usable version specifies an estimative probability from a published scale, a numeric rainfall threshold over a defined duration, the consequence if failure occurs, the confidence level and its basis, and the action and authority that the threshold triggers.

Appendix A: Reusable Templates

A.1 Requirement Development Worksheet

DECISION: _____
DECISION MAKER: _____ DEADLINE: _____
WHAT CHANGES BASED ON THE ANSWER: _____

PIR: _____
 EEI 1: _____
 Indicators: _____
 EEI 2: _____
 Indicators: _____
 EEI 3: _____
 Indicators: _____

TEST: Is it answerable? ____ Deadline-linked? ____
 Would the answer change the decision? ____

A.2 Collection Plan Template

Indicator	Discipline	Specific source	Owner	Cost	Time	Legal / ethical note	Status

A.3 Source Evaluation Quick Reference

Reliability: **A** completely reliable · **B** usually reliable · **C** fairly reliable · **D** not usually reliable · **E** unreliable · **F** cannot be judged

Credibility: **1** confirmed · **2** probably true · **3** possibly true · **4** doubtful · **5** improbable · **6** cannot be judged

A.4 Estimative Language Scale

Almost certainly 95-99% · Very likely 80-95% · Likely 55-80% · Roughly even 45-55% · Unlikely 20-45% · Very unlikely 5-20% · Almost certainly not 1-5%

A.5 Confidence Statement Formula

"We assess with [**high / moderate / low**] **confidence** that [judgment], because [basis: number and independence of sources, assumption load, track record]. This assessment would change if [specific observable]."

A.6 One-Page Product Skeleton

BOTTOM LINE (judgment + confidence + so-what)	3 sentences
KEY FACTS (each rated)	5-8 bullets
ASSESSMENT (meaning + alternative considered)	1-2 paragraphs
KEY ASSUMPTIONS (each with consequence if wrong)	2-4 bullets
INTELLIGENCE GAPS (ranked by analytic impact)	2-4 bullets
RECOMMENDED ACTIONS (specific, assignable, dated)	3-5 bullets
WHAT WOULD CHANGE THIS	2 sentences
COLLECTION REQUESTS	2-3 bullets

Appendix B: Self-Assessment Checklist

Before submitting any product in this chapter, confirm all fifteen:

1. My first sentence contains a judgment.
 2. My first sentence contains a confidence level.
 3. A reader who stops after the first paragraph would act correctly.
 4. Facts are visibly separated from my assessment.
 5. Every fact carries a source and a rating.
 6. I considered at least one alternative explanation and said why I rejected it.
 7. My assumptions are written down.
 8. For each assumption, I said what changes if it is wrong.
 9. My gaps are named, not implied.
 10. My gaps are ranked by how much they would change the assessment.
 11. Every recommendation names a role that can act on it.
 12. Every recommendation has a time frame.
 13. I used estimative language from a published scale, not vague words.
 14. I did not present an assessment as a fact.
 15. Nothing in this product came from a source I obtained improperly.
-

Appendix C: Where to Go Next

Concepts to study after this chapter:

- Structured analytic techniques in depth, particularly ACH, key assumptions checks, and premortem analysis.
- Analytic writing as a discipline of its own, including standards for sourcing, characterization of uncertainty, and distinguishing analysis from opinion.
- Counterintelligence and denial and deception, which is the study of how your own collection can be turned against you.
- Domain specialization: cyber threat intelligence, financial crime analysis, crime analysis, or competitive intelligence, depending on your career direction.
- Legal frameworks relevant to your jurisdiction and sector, including privacy law, trade secret law, and any regulatory obligations governing information handling in your industry.

Professional practice to build now:

- Write one assessment a week about something you care about, using the one-page skeleton in Appendix A.6. Assign a confidence level. Date it. Six months later, score yourself.
 - Keep a forecast log. Analysts who track their own accuracy calibrate better than analysts who do not, and no other habit produces as much improvement for as little effort.
 - Practice writing the opposing case in good faith. It is the hardest skill in this chapter and the one that most separates analysts from advocates.
-

End of Chapter 1